International Journal of Approximate Reasoning 82 (2017) 138–160

Contents lists available at ScienceDirect

International Journal of Approximate Reasoning

www.elsevier.com/locate/ijar

Propositional syntax and semantics induced knowledge re-structuring in a fuzzy logic network for *ad hoc* reasoning

R. Janarthanan [a,*], Amit Konar [a], Aruna Chakraborty [b]

[a] *Artificial Intelligence Lab, ETCE Dept., Jadavpur University, Kolkata-32, India*
[b] *CSE Department, St. Thomas' College of Engineering and Technology, Kolkata-23, India*

ARTICLE INFO

Article history:
Received 14 June 2016
Received in revised form 4 November 2016
Accepted 15 December 2016
Available online 28 December 2016

Keywords:
Propositional syntax and semantics
Fuzzy logic network
Ad hoc reasoning
Rule transformation/re-structuring

ABSTRACT

Traditional approaches to fuzzy reasoning usually employ Generalized Modus Ponens, Generalized Modus Tollens and Generalized Hypothetical Syllogisms to derive fuzzy inferences from a given set of fuzzy observations/facts and fuzzy if-then rules. However, the above approaches occasionally fail to generate inferences, although semantically plausible fuzzy inferences follow as a natural consequence of the given set of rules and fuzzy facts/observations. This paper serves to fill this void. It extends propositional syntax and semantics in the context of fuzzy reasoning to transform/re-structure individual rules to a suitable form (by negating and transposing propositions from antecedent to consequent and vice-versa), so as to get the propositions present in the antecedent (consequent) instantiated (refuted) by positive (negative) facts to enable the rules for forward (backward) firing. Next fuzzy compositional rule of inference is used to perform forward/backward reasoning. Consequently, when the rules are embedded in a fuzzy Petri net like structure, the rules apparently fire randomly in forward or backward direction without maintaining any topological order in transition-firing. Such random firing of rules in distributed parts of the network is referred to as *ad hoc* reasoning. Here, we transform such randomly ordered forward/backward firing sequence of transitions in the network into a fixed topological order of transition-firing in forward direction only by replacing backward firing transitions into equivalent forward firing transitions. The proposed method of automated reasoning has successfully been applied in diagnostic application of an electronic rectifier circuit, and the results are appealing.

© 2016 Elsevier Inc. All rights reserved.

1. Introduction

The logic of fuzzy sets extends propositional calculus [19] from binary to multi-valued truth functionality of propositions. A few well-known fuzzy extensions of propositional counterpart include Generalized Modus Ponens (GMP), Generalized Modus Tollens (GMT) and Generalized Hypothetical Syllogism (GHS) [17], which in propositional logic are referred to as modus ponens, modus tollens and chaining respectively. Most of the fuzzy reasoning problems are designed using the basic framework of GMP, GMT and GHS. However, there exist several fuzzy reasoning problems, which cannot be formulated simply by using the above three reasoning formalisms. Naturally, even if there exist semantically plausible solutions to those problems, existing methods are incompetent to obtain them. This paper attempts to overcome the impasse by ensuring that

* Corresponding author.
E-mail addresses: srmjana_73@yahoo.com (R. Janarthanan), konaramit@yahoo.co.in (A. Konar), aruna.stcet@gmail.com (A. Chakraborty).

http://dx.doi.org/10.1016/j.ijar.2016.12.009
0888-613X/© 2016 Elsevier Inc. All rights reserved.

it would be able to generate fuzzy inferences of a given reasoning problem, in case such inferences follow logically, satisfying both the syntax and the semantics of propositional logic and their fuzzy extensions.

Several formalisms of syntactical and semantic approach to reasoning in propositional logic are found in the literature [1,2]. We here extend propositional forward and backward chaining as the fundamental semantics [1], and the 'negation removal' step of Wang's algorithm, outlined in [1], as the fundamental syntax-preserving rule for automated reasoning in fuzzy logic. It may be recollected that propositional forward chaining asserts that all the propositions in the antecedent being true, at least one of the propositions in the consequent of the rule essentially has to be true. The propositional backward chaining, on the other hand, asserts that at least one of the propositions in the antecedent is false, when all the propositions in the consequent are found false. The principles of forward/backward chaining, introduced above, make sense for a fixed format of propositional rules with premises being ANDed in the antecedent and consequences being ORed in the consequent of the propositional rules [1]. Naturally, introducing the principles of forward/backward reasoning in fuzzy logic with fuzzy production rules (FPRs) or simply rules of the above mentioned structures would have an additional merit to check the semantics of the rules.

In Wang's algorithm, outlined in [1], we replace a rule by an equivalent rule with propositions transposed from the left (right) to the right (left) of the if-then operator after reversing their signs. An extension of this principle in fuzzy logic would enable a traditionally non-fired FPR to fire, if all excluding one fuzzy proposition present in the antecedent (the consequent) of the given rule after transposition are instantiated (refuted) by the available signed observations in the database. It is indeed important to note that semantically such FPRs should be fired, but traditional formalisms (GMP, GMT and GHS) of reasoning in fuzzy logic, are incompetent to serve the purpose. Here lies the importance of the present paper. We here prove that the Diens–Rescher type implication relation of a FPR remains unaltered after transposition of one or more propositions in the rule with complemented sign. We further prove that a properly transformed FPR with all excluding one proposition instantiated by signed fuzzy observations can generate inferences using GMP.

In brief, we would like to adopt propositional semantics to determine the firing conditions of FPRs and then generate fuzzy inference for the desired proposition in the transformed rule using GMP by utilizing the principle of the well-known compositional rule of inference. The latter is referred to as syntactic approach of fuzzy reasoning. The problem ends here for a single FPR with all but one proposition in the rule having proper instantiations. However, there lie more problems when there exists a number of FPRs in the system. What to do exactly, if two or more FPRs are ready for firing? What to do when because of firing of one FPR, the inference generated is used up by another FPR to get ready for firing? All these issues require a controller to take care of these complexities. In this paper, we propose an automated approach to control the intricacies required before and after firing of a FPR. The arrangement required before firing of a FPR is to make it ready for firing by necessary transposition of propositions, so that all excluding one proposition of the FPR are instantiated with available facts. The intricacy required after firing of a FPR is to transfer the generated inference to the next FPR waiting for instantiation of one of its premise or consequent. Such transfer of inference from one fired FPR to a second can be easily realized, if the FPRs are connected by a network having shared propositions between antecedent (consequent) of the first FPR and consequent (antecedent) of a second FPR.

It may be noted that there is no fixed pre-order of FPR firing in the network. The firing order of FPRs and nature of firing (forward/backward) are dictated by the arrival of instantiations of a FPR from others connected with the former FPR. Since such arrival is apparently random in a big network of several hundreds to thousand FPRs, the rules apparently look like fired (forward/backward) randomly over time. Such firing of FPRs in the network without having a fixed pre-order of firing for inference generation has been referred to as *ad hoc* reasoning [22]. It is indeed important to mention here that the word *ad hoc* has similarity with the same phrase in the context of communication networks. It may be recollected that in an *ad hoc* network the connections are established apparently randomly between two network nodes (stations) without any centralized administration [52]. Although, in the present context, a FPR is connected with one or more FPRs in the network through shared propositions, the active data pathway (between two rules) for inference transfer is determined only after firing of a FPR. The phrase *ad* hoc reasoning used in the title thus makes sense.

Although several topologies of the network may support *ad hoc* reasoning, we because of the flexibility of knowledge representation and reasoning capability of Fuzzy Petri Nets (FPNs) [3–8,21–51], coupled with our past experience of using such topology [7,21], prefer a structure graphically similar to Petri nets [9]. Both forward and backward reasoning models of FPNs are available in the literature. Among the forward reasoning models of FPNs, the works by Chen et al. [4], Gao et al. [31], Liu et al. [49,50] need special mention. In [4], Chen et al. proposed an efficient algorithm of reasoning on a FPN using reachability analysis. Gao et al. in [31] introduced a novel scheme of fuzzy reasoning using the operators in max-algebra. They adopted a matrix equation format, similar to that used in traditional Petri nets [9]. Moreover, they indicated one approach to represent negation in the FPN model. In [49], Liu et al. proposed a linguistic reasoning Petri net model to design an ordered weighted linguistic reasoning algorithm for knowledge representation and reasoning. Very recently Liu et al. proposed a new model of FPN [50] using intuitionistic fuzzy sets and ordered weighted averaging operators, and developed intuitionistic fuzzy reasoning algorithm using max-algebra. Among the backward reasoning models of FPNs used for diagnostic applications, the works by Hu et al. [30] and Liu et al. [51] need special mention. In [30], Hu et al. presented a model of reversed FPN to analytically describe the knowledge of a diagnostic problem in manufacturing systems, and also proposed an iterative algorithm using max-algebra to deduce the relationship between the manifestation and the cause. Recently, Liu et al. in [51] proposed a novel model of fault diagnosis and cause analysis for abnormal events using fuzzy evidential reasoning and dynamic adaptive FPNs. Among the other significant works on FPN, high level reasoning model by

Scarpelli et al. [14], concurrent reasoning model by Wang et al. [15], predicate-transition model by Yu [16], and supervised [11,24] and unsupervised learning [25] applications of FPN models need mentioning.

The novelty of the present work with respect to the existing works on FPN or in general fuzzy logic networks (FLNs) are summarized below. First, the authors here extend propositional reasoning to restructure individual FPRs by signed transposition of the propositions from the antecedent to the consequent and vice-versa with an aim to have possible instantiations (refutations) for the propositions present in the antecedent (consequent). Consequently, semantically friable FPRs, which occasionally do not fire in traditional fuzzy reasoning, can fire now with the introduced formalisms. Second, we transform a mixed forward/backward fire-able reasoning system into an equivalent forward fire-able system that produces the same inferences as before. Third, the proposed reasoning algorithm has been proved to be both sound and semantically complete, where most of the existing techniques ensure soundness of inferences but lack in completeness. Finally, we here demonstrate the possible application of the proposed *ad hoc* reasoning system in a diagnostic problem, where both causes and effects of the faults, determine the faults by a special type of fuzzy reasoning as undertaken in probabilistic reasoning framework by Pearl [10,53] long before. However, the probabilistic reasoning system requires a number of conditional probabilities, which are not easily available, whereas a subject expert assigns the membership functions to fuzzy propositions intuitively, thus justifying the essence of the proposed reasoning technique. The paper is an extension of the authors' previous work [22]. The extension includes algorithm developments and proof of soundness and completeness, whereas in [22] we simply illustrated the principles without detailed algorithms and analysis.

The paper is divided into eight sections. In Section 2, we extend propositional semantics, such as forward and backward chaining, in the framework of a fuzzy reasoning system. Here, we also prove one fundamental syntactic property of propositional logic regarding transfer of one or more sign-inverted propositional symbol to the other side of an implication rule, which is commonly used in Wang's algorithm. Next we extend this property for fuzzy reasoning, and lastly we attempt to combine syntax and semantics for fuzzy reasoning. In Section 3, we define fuzzy logic network and also propose an algorithm for *ad hoc* reasoning by composite use of syntax and semantics as outlined in Section 2. Here, we also illustrate the *ad hoc* reasoning algorithm with a given fuzzy logic network. Section 4 introduces a general scheme to transform an *ad hoc* reasoning problem into an equivalent forward firing model. The algorithm for forward reasoning introduced in Section 4 is illustrated in Section 5 with respect to a diagnostic problem of an electronic rectifier circuit. Soundness and completeness issues of an *ad hoc* reasoning problem, transformed into an equivalent forward reasoning problem, are discussed in Section 6. A comparative study of the present work with relevant existing works is given in section 7. Conclusions are listed in Section 8.

2. Propositional extension for fuzzy reasoning

This section includes three main sub-sections. Section 2.1 deals with fuzzy extension of propositional semantics. Section 2.2 deals with fuzzy extension of propositional syntax, and section 2.3 offers a general method of fuzzy reasoning with a single FPR using the said extensions.

2.1. Fuzzy extension of propositional semantics

This section extends the propositional semantics for its suitability in fuzzy reasoning. In propositional logic, we use forward and backward chaining [1], reproduced below for reasoning using semantic approach. Given a propositional rule

$$\text{R: } p_1 \wedge p_2 \wedge \ldots . \wedge p_n \rightarrow q_1 \vee q_2 \vee \cdots \vee q_m,$$

where the propositions $p_1, p_2, \ldots, p_n$ in the antecedent are connected by AND ($\wedge$) operators, the propositions $q_1, q_2, \ldots, q_m$ in the consequent are connected by OR ($\vee$) operators and $\rightarrow$ denotes a if-then operator. Now, in case $p_1, p_2, \ldots, p_n$ all are true then at least one of q_1 or q_2 or…or q_m is true. The last statement is the test criterion of *forward chaining*. Similarly, in case all the propositions: $q_1, q_2, \ldots, q_m$ listed in the consequent are false, then at least one of the premises: p_1 or p_2 or…or p_n is false. This is referred to as propositional *backward chaining*. A propositional rule, also called a theorem, is semantically correct, if both forward and backward chaining hold jointly [1]. We now extend propositional forward and backward chaining principles for inference generation.

Propositional extensions: The following two extensions semantically follow from the definitions of forward and backward chaining.

1. *It is apparent from forward chaining that if all the propositions in the antecedent are true and all but one proposition in the consequent is false then the remaining proposition in the consequent is essentially true.*
2. *If all the propositions in the consequent are false and all excluding one proposition in the antecedent are true then the last proposition in the antecedent is essentially false.*

Proposed semantic rules for fuzzy forward and backward chaining

A FPR contains fuzzy propositions, having multi-valued truth function (membership function lying in [0, 1]), in both the antecedent and the consequent of the rule. Firing of FPRs satisfying forward/backward chaining requires partial/approximate matching of the fuzzy propositions of the antecedent and the consequent with fuzzy observations. Given a FPR: If P_1 and

P_2 and ... and P_n Then Q_1 or Q_2 or ... or Q_m where $P_i = x_i$ is A_i and $Q_j = y_j$ is B_j are fuzzy propositions and $x_i \in X_i$ for $i = 1$ to n and $y_j \in Y_j$ for $j = 1$ to m are linguistic variables, and A_i and B_j are fuzzy sets in the universes X_i and Y_j respectively. We now use propositional extensions introduced above to develop semantic rules for fuzzy forward and backward reasoning.

To explain fuzzy forward chaining, let $P_i' = x_i$ is A_i' for $i = 1$ to n be fuzzy observation, where A_i' is a fuzzy set semantically (meaningfully) close to the fuzzy set A_i [17,18]. The more intuitively closer A_i' is with A_i, the larger is the membership $\mu_{A_i'}(x_i)$. This is referred to as partial/approximate instantiation of $P_i = x_i$ is A_i in the given rule by the observation x_i is A_i'. Let us also assume that the observation Q_j', which is an approximation of Q_j, partially/approximately refutes proposition Q_j, present in the consequent for $j = 1$ to m, $j \neq k$. In other words, Not Q_j' is instantiated by Q_j, for all j. Then Q_k', an approximation of Q_k, *would be the fuzzy inference*.

Semantic rule for fuzzy forward chaining: *If the fuzzy proposition P_i for all i in the antecedent of a FPR is partially/approximately instantiated by the available fuzzy observations P_i' and proposition Q_j for all j, excluding one proposition Q_k in the consequent is partially/approximately refuted by the fuzzy observations Q_j' (or supported by* Not Q_j'*) then the inference Q_k' generated following an approximation of the remaining proposition Q_k in the consequent would be partially true with a membership in* $[0, 1]$.

Now, to explain fuzzy backward chaining, let the fuzzy proposition Q_j in the consequent of the FPR be approximately/partially refuted by observation Q_j' for all j, and proposition P_i for all i, excluding the k-th fuzzy proposition in the antecedent is partially instantiated by the observation P_i'. Then the approximation of the k-th fuzzy proposition P_k' in the antecedent would be partially false with a degree of membership in $[0, 1]$. In other words, P_k' would be semantically complement to P_k.

Semantic rule for fuzzy backward chaining: *If the fuzzy proposition Q_j for all j in the consequent of a FPR is partially/approximately refuted by observation Q_j' (or, supported by* Not Q_j'*) and proposition P_i for all i, excluding the fuzzy proposition P_k in the antecedent, is partially instantiated by the available observation P_i' then the inference P_k' generated following an approximation of the proposition P_k, would be partially false with a membership in* $[0, 1]$.

Example 1. Let us consider the FPR: *If height is TALL and weight is MODERATE then speed is HIGH or respiration-rate is HIGH*. Here, height, weight, speed and respiration-rate are linguistic variables and TALL, MODERATE and HIGH are fuzzy sets in respective universes. The measured MFs of all the four fuzzy propositions are given. Also given the observed MFs of weight is MORE-OR-LESS-MODERATE, speed is NOT-MORE-OR-LESS-HIGH and respiration rate is NOT-MORE-OR-LESS-HIGH, then we by fuzzy backward chaining infer that height is NOT-MORE-OR-LESS-TALL. Here, we would use the measured MFs to compute implication relation for the given rule, and observed MFs to determine the inference by using max-min compositional rule of inference.

Similarly, given the observed MFs of height is MORE-OR-LESS-TALL, weight is MORE-OR-LESS-MODERATE and speed is NOT-MORE-OR-LESS-HIGH, we infer by fuzzy forward reasoning that the MF of respiration-rate is MORE-OR-LESS-HIGH. The calculation of the inferred MF will be taken up in the next section. It may be noted here that during instantiation of a FPR by observed facts, we allow partial matching of the propositions. For example, although the FPR contains 'height is TALL' in the antecedent, we allow an instantiation of height is MORE-OR-LESS-TALL as the observed data/fact.

2.2. Extension of propositional syntax for fuzzy reasoning

Let p_i and q_j for $i = 1$ to n and $j = 1$ to m be binary propositions present respectively in the antecedent and the consequent of a propositional rule of the following form:

$$\text{Rule-1: } p_1 \wedge p_2 \wedge \ldots \wedge p_n \rightarrow q_1 \vee q_2 \vee \ldots \vee q_m,$$

where $\rightarrow$ denotes the if-then (implication) operator, and $\wedge$ in the left and $\vee$ in the right hand side of the if-then operator represent AND (conjunction) and OR (disjunction) operators respectively. Theorem 1 below provides a property about transformation of the above rule to an equivalent rule with one or more propositions transposed to the other side of the if-then operator after complementation.

Theorem 1 *(Propositional transposition). One or more propositional symbols, if transposed in Rule-1 to the other side of the If-Then operator, get connected in complemented form with the exiting antecedent through conjunction and with existing consequent through disjunction operators.*

Proof. To prove the statement, let us transpose p_1 through p_r and q_1 through q_s to the other side of the if-then operator by substitution of standard propositional theorems [1]. Now, Rule-1 is transformed to the desired form by the following steps.

$$
\begin{aligned}
&p_1 \wedge p_2 \wedge p_3 \wedge \ldots \wedge p_n \rightarrow q_1 \vee q_2 \vee \ldots \vee q_m \\
&\quad \equiv \neg(p_1 \wedge p_2 \wedge p_3 \wedge \ldots \wedge p_n) \vee (q_1 \vee q_2 \vee \ldots \vee q_m) \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{(Since } p \rightarrow q \equiv \neg p \vee q) \\
&\quad \equiv (\neg p_1 \vee \neg p_2 \vee \ldots \vee \neg p_n) \vee (q_1 \vee q_2 \vee \ldots \vee q_m) \quad \text{(By De Morgan's law)} \\
&\quad \equiv \left\{(\neg p_{r+1} \vee \neg p_{r+2} \vee \ldots \vee \neg p_n) \vee (q_1 \vee q_2 \vee \ldots \vee q_s)\right\} \vee \\
&\qquad \left\{(\neg p_1 \vee \neg p_2 \vee \ldots \vee \neg p_r) \vee (q_{s+1} \vee q_{s+2} \vee \ldots \vee q_m)\right\} \\
&\quad \equiv \left\{\neg(\neg q_1 \wedge \neg q_2 \wedge \ldots\ldots \wedge \neg q_s) \vee (\neg p_{r+1} \vee \neg p_{r+2} \vee \ldots\ldots \vee \neg p_n)\right\} \\
&\qquad \vee \left\{(\neg p_1 \vee \neg p_2 \vee \ldots\ldots \vee \neg p_r) \vee (q_{s+1} \vee q_{s+2} \vee \ldots\ldots \vee q_m)\right\} \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{(By De Morgan's law)} \\
&\quad \equiv \left[\neg(\neg q_1 \wedge \neg q_2 \wedge \ldots\ldots \wedge \neg q_s) \vee \neg\left\{(p_{r+1} \wedge p_{r+2} \wedge \ldots\ldots \wedge p_n)\right\}\right] \\
&\qquad \vee \left[(\neg p_1 \vee \neg p_2 \vee \ldots\ldots \vee \neg p_r) \vee (q_{s+1} \vee q_{s+2} \vee \ldots\ldots \vee q_m)\right] \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{(By De Morgan's law)} \\
&\quad \equiv \neg\left[(\neg q_1 \wedge \neg q_2 \wedge \ldots\ldots \wedge \neg q_s) \wedge (p_{r+1} \wedge p_{r+2} \wedge \ldots\ldots \wedge p_n)\right] \\
&\qquad \vee \left\{(\neg p_1 \vee \neg p_2 \vee \ldots\ldots \vee \neg p_r) \vee (q_{s+1} \vee q_{s+2} \vee \ldots\ldots \vee q_m)\right\} \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{(By De Morgans law)} \\
&\quad \equiv \neg q_1 \wedge \neg q_2 \wedge \ldots \wedge \neg q_s \wedge p_{r+1} \wedge p_{r+2} \wedge \ldots \wedge p_n \rightarrow \neg p_1 \vee \neg p_2 \vee \ldots\ldots \vee \neg p_r \\
&\qquad \vee q_{s+1} \vee q_{s+2} \vee \ldots \vee q_m \qquad\qquad\qquad\qquad\qquad\qquad (\because \neg p \vee q \equiv p \rightarrow q)
\end{aligned}
$$

Since the above identity holds for any $r \in [0, n]$ number of propositions in the antecedent and any $s \in [0, m]$ number of propositions in the consequent of Rule-1, the theorem holds. □

We now introduce the cumulative t-norm and s-norm operators [12] to discuss the rest of the paper.

Let $P_i = x_i$ is A_i and $P_j = x_j$ is A_j be two fuzzy propositions, where $x_i \in X_i = \{x_i^1, x_i^2, ..., x_i^m\}$ and $x_j \in X_j = \{x_j^1, x_j^2, ..., x_j^n\}$; A_i and A_j are two fuzzy sets in respective domains X_i and X_j. Let $\mu_{A_i}(x_i)$ and $\mu_{A_j}(x_j)$ be the membership functions (MFs) of x_i is A_i and x_j is A_j respectively in $[0, 1]$. Now, we define s-norm relation between two membership functions.

$$
\begin{aligned}
&\mu_{A_i}(x_i)\, \mathrm{s}\, \mu_{A_j}(x_j) \\
&= \left\{\mu_{A_i}(x_i^1)\, \mathrm{s}\, \mu_{A_j}(x_j^1), \mu_{A_i}(x_i^2)\, \mathrm{s}\, \mu_{A_j}(x_j^1), ..., \mu_{A_i}(x_i^m)\, \mathrm{s}\, \mu_{A_j}(x_j^1),\right. \\
&\quad \mu_{A_i}(x_i^1)\, \mathrm{s}\, \mu_{A_j}(x_j^2), \mu_{A_i}(x_i^2)\, \mathrm{s}\, \mu_{A_j}(x_j^2), ..., \mu_{A_i}(x_i^m)\, \mathrm{s}\, \mu_{A_j}(x_j^2), \\
&\quad \ldots\ldots \quad \ldots\ldots \quad \ldots\ldots \quad \ldots\ldots \quad \ldots\ldots \quad \ldots\ldots \\
&\quad \left.\mu_{A_i}(x_i^1)\, \mathrm{s}\, \mu_{A_j}(x_j^n), \mu_{A_i}(x_i^2)\, \mathrm{s}\, \mu_{A_j}(x_j^n), ..., \mu_{A_i}(x_i^m)\, \mathrm{s}\, \mu_{A_j}(x_j^n)\right\}
\end{aligned}
$$

Since $|X_i| = m$ and $|X_j| = n$, $P_i\, \mathrm{S}\, P_j$ has $m \times n$ elements. Further let, $P_k = x_k$ is A_k and $x_k \in X_k = \{x_k^1, x_k^2, ..., x_k^l\}$. Then for $i = 1$, $j = 2$ and $k = 3$, we introduce cumulative s-norm

$$
\begin{aligned}
&\overset{3}{\underset{i=1}{\mathrm{S}}}\left(\mu_{A_i}(x_i)\right) \\
&= \left\{\mu_{A_1}(x_1)\, \mathrm{s}\, \mu_{A_2}(x_2)\, \mathrm{s}\, \mu_{A_3}(x_3) : x_1 \in X_1, x_2 \in X_2, x_3 \in X_3\right\} \\
&= \left\{\mu_{A_1}(x_1)\, \mathrm{s}\, \mu_{A_2}(x_2)\right\}\, \mathrm{s}\, \left\{\mu_{A_3}(x_3)\right\} \\
&= \left\{\overset{2}{\underset{i=1}{\mathrm{S}}}\left(\mu_{A_i}(x_i)\right)\right\}\, \mathrm{s}\, \left\{\mu_{A_3}(x_3)\right\},
\end{aligned}
$$

which will have $m \times n \times l$ elements. Similarly, we define cumulative t-norm

$$
\overset{3}{\underset{i=1}{\mathrm{T}}}\left(\mu_{A_i}(x_i)\right) = \left\{\overset{2}{\underset{i=1}{\mathrm{T}}}\left(\mu_{A_i}(x_i)\right)\right\}\, \mathrm{t}\, \left\{\mu_{A_3}(x_3)\right\},
$$

where t denotes the t-norm operator. Defining $x_i \in X_i$ and fuzzy set A_i in universe X_i we, in general, can express

$$
\begin{aligned}
&\overset{n}{\underset{i=1}{\mathrm{T}}}\left(\mu_{A_i}(x_i)\right) \\
&= \left\{\mu_{A_1}(x_1)\, \mathrm{t}\, \mu_{A_2}(x_2)\, \mathrm{t} \ldots \mathrm{t}\, \mu_{A_{n-1}}(x_{n-1})\right\}\, \mathrm{t}\, \left\{\mu_{A_n}(x_n)\right\} \\
&= \left\{\overset{n-1}{\underset{i=1}{\mathrm{T}}}\left(\mu_{A_i}(x_i)\right)\right\}\, \mathrm{t}\, \left\{\mu_{A_n}(x_n)\right\}.
\end{aligned}
$$

Similar expression can also be obtained by replacing t-norm and T-norm by s-norm and S-norm operators. Now, we like to extend Theorem 1 in the context of fuzzy propositional rules.

Theorem 2 *(Fuzzy transposition). Given a FPR: P_1 and P_2 and ... and $P_n \rightarrow Q_1$ or Q_2 or ... or Q_m, where $P_i = x_i$ is A_i and $Q_j = y_j$ is B_j for $i = 1$ to n, and $j = 1$ to m. If any r number of P_is and s number of Q_js are negated (complemented) and transposed to the other side of the implication to get connected with the existing consequent by OR-operator and with the existing antecedent by AND-operator respectively, then the resulting rule will be equivalent to the original rule by Diens–Rescher implication function.*

Proof. Let $\mu_{A_i}(x_i)$ and $\mu_{B_j}(y_j)$ denote the membership functions for x_i is A_i and y_j is B_j respectively. Then for the given FPR, the implication relation obtained by Diens–Rescher implication function [20] is given by

$$\begin{aligned}
&R_1(x_1, x_2, ..., x_n; y_1, y_2, ..., y_m)\\
&\quad = c\big(\mu_{A_1}(x_1)\,\mathrm{t}\,\mu_{A_2}(x_2)\,\mathrm{t}\,...\,\mathrm{t}\,\mu_{A_n}(x_n)\big)\,\mathrm{s}\,\big(\mu_{B_1}(y_1)\,\mathrm{s}\,\mu_{B_2}(y_2)\,\mathrm{s}\,...\,\mathrm{s}\,\mu_{B_m}(y_m)\big)\\
&\qquad\qquad\qquad

and

$$R_l(x_1, x_2, ..., x_n, y_1, y_2, ..., y_{l-1}, y_{l+1}, ..., y_m; y_l),$$

are fuzzy implication relations obtained by Diens–Rescher implication function and '∘' denotes max-min composition operator [12].

Proof. The proof has two parts. For the first part (fuzzy backward reasoning), we have $\mu_{\bar{A}'_k}(x_k)$ as the inference. We first re-organize the given FPR in the format:

$$P_1 \wedge P_2 \wedge ... \wedge P_{k-1} \wedge P_{k+1} \wedge ... \wedge P_n \wedge \bar{Q}_1 \wedge \bar{Q}_2 \wedge ... \wedge \bar{Q}_m \rightarrow \bar{P}_k.$$

The implication relation for the FPR using t-norm is obtained as

$$\begin{aligned} &R_k(x_1, x_2, ..., x_{k-1}, x_{k+1}, ..., x_n, y_1, y_2, ..., y_m; x_k) \\ &= \left[\left\{1 - \left\{\left(\mu_{A_1}(x_1) \,\mathrm{t}\, \mu_{A_2}(x_2) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{A_{k-1}}(x_{k-1}) \,\mathrm{t}\, \mu_{A_{k+1}}(x_{k+1}) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{A_n}(x_n)\right) \mathrm{t}\right.\right.\right. \\ &\quad \left.\left.\left.\left(\mu_{\bar{B}_1}(y_1) \,\mathrm{t}\, \mu_{\bar{B}_2}(y_2) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{\bar{B}_m}(y_m)\right)\right\}\right\} \mathrm{s} \left\{\mu_{\bar{A}_k}(x_k)\right\}\right] \\ &= \left[\left\{1 - \left\{ \underset{\substack{i=1,\\ i\neq k}}{\overset{n}{\mathrm{T}}} \left(\mu_{A_i}(x_i)\right)\right\} \mathrm{t} \left\{ \underset{j=1}{\overset{m}{\mathrm{T}}} \left(\mu_{\bar{B}_j}(y_j)\right)\right\}\right\} \mathrm{s} \left\{\mu_{\bar{A}_k}(x_k)\right\}\right]_{(w^{n+m-1} \times w)} \end{aligned}$$

and the firing strength of the FPR is obtained as

$$\begin{aligned} [\mu_{(antecedent)}] = &\left[\left(\mu_{A'_1}(x_1) \,\mathrm{t}\, \mu_{A'_2}(x_2) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{A'_{k-1}}(x_{k-1}) \,\mathrm{t}\, \mu_{A'_{k+1}}(x_{k+1}) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{A'_n}(x_n)\right) \mathrm{t}\right. \\ &\left.\left(\mu_{\bar{B}'_1}(y_1) \,\mathrm{t}\, \mu_{\bar{B}'_2}(y_2) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{\bar{B}'_m}(y_m)\right)\right]_{(1 \times w^{n+m-1})} \end{aligned}$$

where MF $\mu_{A'_i}(x_i)$ and $\mu_{B'_j}(y_j)$ are represented by w dimensional vectors, and the inference is obtained by the following two steps:

$$\left[\mu_{\bar{A}'_k}(x_k)\right] = [\mu_{(antecedent)}] \circ R_k.$$

Thus, $\mu_{A'_k}(x_k) = 1 - \mu_{\bar{A}'_k}(x_k)$.

For the second part (fuzzy forward reasoning) of the proof, we have $\mu_{B'_l}(y_l)$ as the inference. We have restructured the rule as follows:

$$\text{If } (P_1 \text{ and } P_2 \text{ and } ... \text{ and } P_n \text{ and } \bar{Q}_1 \text{ and } \bar{Q}_2 \text{ and } ... \text{ and } \bar{Q}_{l-1} \text{ and } \bar{Q}_{l+1} \text{ and } ... \text{ and } \bar{Q}_m) \text{ Then } Q_l.$$

The implication relation for the FPR is obtained as

$$\begin{aligned} &R_l(x_1, x_2, ..., x_n, y_1, y_2, ..., y_{l-1}, y_{l+1}, ..., y_m; y_l) \\ &= \left[\left\{1 - \left\{\left(\mu_{A_1}(x_1) \,\mathrm{t}\, \mu_{A_2}(x_2) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{A_n}(x_n)\right) \mathrm{t}\right.\right.\right. \\ &\quad \left.\left.\left.\left(\mu_{\bar{B}_1}(y_1) \,\mathrm{t}\, \mu_{\bar{B}_2}(y_2) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{\bar{B}_{l-1}}(y_{l-1}) \,\mathrm{t}\, \mu_{\bar{B}_{l+1}}(y_{l+1}) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{\bar{B}_m}(y_m)\right)\right\}\right\}\right] \mathrm{s} \left(\mu_{B_l}(y_l)\right) \\ &= \left[\left\{1 - \left\{ \underset{i=1}{\overset{n}{\mathrm{T}}} \left(\mu_{A_i}(x_i)\right)\right\} \mathrm{t} \left\{ \underset{\substack{j=1,\\ j\neq l}}{\overset{m}{\mathrm{T}}} \left(\mu_{\bar{B}_j}(y_j)\right)\right\}\right\} \mathrm{s} \left(\mu_{B_l}(y_l)\right)\right]_{(w^{n+m-1} \times w)} \end{aligned}$$

and the firing strength of the FPR is obtained as

$$\begin{aligned} [\mu_{(antecedent)}] = &\left[\left(\mu_{A'_1}(x_1) \,\mathrm{t}\, \mu_{A'_2}(x_2) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{A'_n}(x_n)\right) \mathrm{t}\right. \\ &\left.\left(\mu_{\bar{B}'_1}(y_1) \,\mathrm{t}\, \mu_{\bar{B}'_2}(y_2) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{\bar{B}'_{l-1}}(y_{l-1}) \,\mathrm{t}\, \mu_{\bar{B}'_{l+1}}(y_{l+1}) \,\mathrm{t}\, ... \,\mathrm{t}\, \mu_{\bar{B}'_m}(y_m)\right)\right]_{(1 \times w^{n+m-1})} \end{aligned}$$

and the inference,

$$\left[\mu_{B'_l}(y_l)\right] = [\mu_{(antecedent)}] \circ R_l.$$

Thus the theorem follows. □

Note. In case the MF of $\mu_{\bar{B}_j}(y_j)$ is not known but the MF of $\mu_{B_j}(y_j)$ is known, we obtain $\mu_{\bar{B}_j}(y_j) = 1 - \mu_{B_j}(y_j)$, $\exists j$ as required.

2.3. Combining syntax and semantics for inference generation from a given rule

This section proposes a more general form of fuzzy reasoning, where all excluding only one fuzzy proposition in a FPR have observed MFs. It is also assumed that measured MFs of all fuzzy propositions are available, and all the propositional symbols used in the antecedent and the consequent sides of the FPRs are positive, such as x is A, although the facts may be negative, such as x is $\bar{A}'$ or positive x is A'. The principles used to handle the problem are briefly outlined below.

1. For each FPR with n antecedent propositions and m consequent propositions, keep the proposition p without observed MF only at the consequent side of the rule with proper sign. If the proposition p has been residing in the consequent side of the FPR, it should be positive. If the proposition p has been residing in the antecedent side, it should be complemented while being placed in the consequent side (by Theorem 2). All other propositions present in the rule excluding proposition p should be placed in the antecedent side with proper sign, i.e., positive sign if transposition is not required and negative sign, otherwise.
2. Construct fuzzy implication relation for the FPR with one proposition p in the consequent and the remaining propositions in the antecedent with proper sign as obtained in step 1 by using the measured MFs of all the propositions in the rule.
3. Derive the MF of the proposition p in the consequent by using the observed MFs of the propositions in the antecedent of the transformed rule and the implication relation for the FPR obtained in step 2 using compositional rule of inference [20] (by Theorem 3).
4. If the proposition in the consequent is negative, complement its MF to obtain the inference. Otherwise, declare the MF obtained in the last step as the inference (by Theorem 3).

The principles narrated above support both semantic and syntactic methods introduced earlier. Theorem 2 is utilized for re-structuring of a given FPR with an aim to instantiate all excluding one fuzzy proposition of the rule by the available facts. So, Theorem 2 ensures semantics checking for forward/backward firing. Theorem 3 is used next for the syntactical part of fuzzy reasoning, which is required to derive fuzzy inferences.

Example 2. Suppose we have a FPR: If P_1 and P_2 then Q_1 or Q_2, where $P_1 = x_1$ is A_1, $P_2 = x_2$ is A_2, $Q_1 = y_1$ is B_1 and $Q_2 = y_2$ is B_2 with usual meaning of the parameters as introduced before. Suppose, we have measured MFs for P_1, P_2, Q_1 and Q_2, and also we have observed MFs of P_1', Not Q_1' and Not Q_2', we need to evaluate the MF of P_2'.
To handle this problem, we first transform the FPR using step-1. The resulting FPR is given by

If P_1 and not Q_1 and not Q_2 Then not P_2.

We then by step-2 determine the fuzzy implication relation for the above FPR, given by

$$R(x_1, y_1, y_2; x_2) = \left[1 - \left(\mu_{A_1}(x_1) \,\mathrm{t}\, \mu_{\bar{B}_1}(y_1) \,\mathrm{t}\, \mu_{\bar{B}_2}(y_2)\right)\right) \mathrm{s} \left(\mu_{\bar{A}_2}(x_2)\right)\right].$$

Next by step 3, we evaluate the fuzzy inference

$$\left[\mu_{\bar{A}_2'}(x_2)\right] = \left[\left(\mu_{A_1'}(x_1) \,\mathrm{t}\, \mu_{\bar{B}_1'}(y_1) \,\mathrm{t}\, \mu_{\bar{B}_2'}(y_2)\right)\right] \circ R.$$

We finally obtain

$$\mu_{A_2'}(x_2) = 1 - \mu_{\bar{A}_2'}(x_2).$$

Example 3. Consider the same FPR as indicated in Example 2 and presume like Example 2 that the measured MFs for P_1, P_2, Q_1 and Q_2 are known, while observed MFs of P_1', P_2' and Q_1' are known, and we need to infer the MF for Q_2'.
To solve the problem, we by step-1 modify the rule as

If P_1 and P_2 and not Q_1 Then Q_2.

In step 2, we compute the implication relation of the FPR:

$$R(x_1, x_2, y_1; y_2) = \left[\left(1 - \left(\mu_{A_1}(x_1) \,\mathrm{t}\, \mu_{A_2}(x_2) \,\mathrm{t}\, \mu_{\bar{B}_1}(y_1)\right)\right) \mathrm{s} \left(\mu_{B_2}(y_2)\right)\right].$$

Next by step 3 we obtain the fuzzy inference

$$\left[\mu_{B_2'}(y_2)\right] = \left[\left(\mu_{A_1'}(x_1) \,\mathrm{t}\, \mu_{A_2'}(x_2) \,\mathrm{t}\, \mu_{\bar{B}_1'}(y_1)\right)\right] \circ R.$$

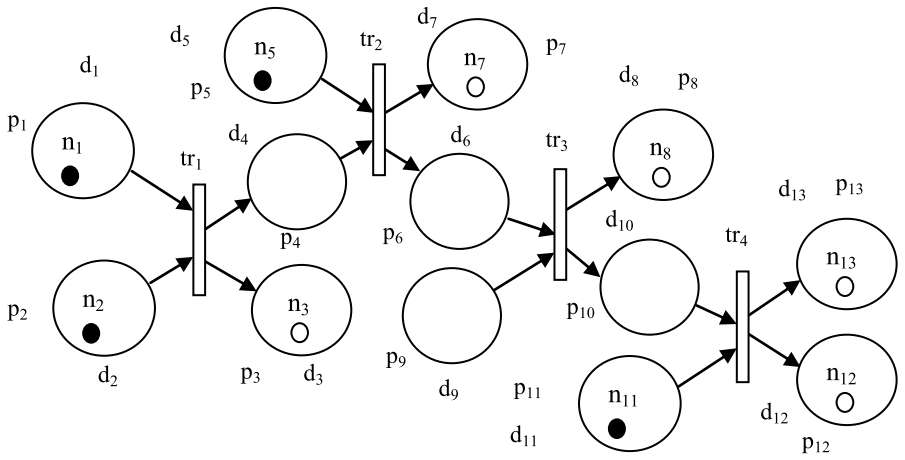


Fig. 1. An FLN used to illustrate its parameters.

3. Automated reasoning in a fuzzy logic network

This section proposes a new algorithm for automated reasoning by a structured approach realized with a fuzzy logic network (FLN). The proposed network attempts to satisfy Theorems 2 and 3, introduced in Section 2. We present below a few definitions followed by an algorithm for automated reasoning using FLN.

Definition 1. An FLN is a directed bipartite graphs with 8-tuples given by $\langle P, D, \mathrm{Tr}, I, O, R_{ij}, N, M\rangle$, where

$P = \{p_1, p_2, \ldots, p_N\}$ denotes a finite set of places,

$D = \{d_1, d_2, \ldots, d_N\}$ represents a set of fuzzy propositions, where proposition d_i has association with place p_i, for $i = 1$ to N,

$\mathrm{Tr} = \{\mathrm{tr}_1, \mathrm{tr}_2, \ldots, \mathrm{tr}_M\}$ is a set of transitions, such that $P \times \mathrm{Tr}$ and $\mathrm{Tr} \times P$ are valid arcs (or flow relations),

I: $\mathrm{Tr} \to P$ represents a mapping function from each transition to its input places. For example, for any i, j, if $p_i \times \mathrm{tr}_j \in \{P \times \mathrm{Tr}\}$ then p_i is an input place of transition tr_j. Symbolically, $p_i \in I(\mathrm{tr}_j)$.

O: $\mathrm{Tr} \to P$ represents a mapping function from each transition to its output places. For instance, for any i, j, if $\mathrm{tr}_j \times p_i \in \{\mathrm{Tr} \times P\}$ then p_i is an output place of transition tr_j. Symbolically, $p_i \in O(\mathrm{tr}_j)$.

R_{ij}: $\mathrm{tr}_i \times p_j \to [0, 1] \times [0, 1]$ represents fuzzy relational matrix R_{ij} at an arc connected between transition tr_i to place p_j, where $p_j \in O(tr_i)$.

N': $D \to [0, 1]^l$ is an association function, representing a mapping from fuzzy propositions to a (membership) vector of l dimension with elements in $[0, 1]$; For any fuzzy proposition $d_i = x_i$ is A_i, we define $N' = [n_i]_{(1\times l)}$, where $n_i = \mu_{A_i}(x_i)$ denotes a membership value in $[0, 1]$ for a specific value of x_i;

M: $P \to \{+1, -1, 0\}$ is an association function, representing tokens at places. We use three possible types of tokens for a given proposition d_i at place p_i : $+1$ to describe that d_i is true, -1 to describe that d_i is false and 0 to represent that truth value of d_i is unknown until the present time. In the graphical model of FLN, we represent $+1$ by solid dots, -1 by void dots and 0 by no dots.

Example 4. In this example, we illustrate the parameters of a FLN. Let $P = \{p_1, p_2, \ldots, p_{13}\}$, $\mathrm{Tr} = \{\mathrm{tr}_1, \mathrm{tr}_2, \mathrm{tr}_3, \mathrm{tr}_4\}$, $D = \{d_1, d_2, d_3, \ldots, d_{13}\}$, where $d_i = x_i$ is A_i is a fuzzy proposition for $i = 1$ to 13. $R_{1,3}$, $R_{1,4}$, $R_{2,6}$, $R_{2,7}$, $R_{3,8}$, $R_{3,10}$, $R_{4,12}$, $R_{4,13}$ are fuzzy relations of the form $R_{i,j}$, representing connectivity from transition tr_i place p_j. Here, $N' = \{n_1, n_2, ..., n_{13}\}$ with n_i for $i = 1$ to 13, where n_i is associated with proposition d_i at place p_i.

In Fig. 1, we assign tokens at 9 places with solid dots at places p_1, p_2, p_5, and p_{11} and void dots at places p_3, p_7, p_8, p_{12} and p_{13} and no dots at places p_4, p_6, p_9, and p_{10}. Formally, $M(p_1) = +1$, $M(p_2) = +1$, $M(p_5) = +1$, $M(p_{11}) = +1$, the corresponding MFs are n_1, n_2, n_5 and n_{11}. Similarly, $M(p_3) = -1$, $M(p_7) = -1$, $M(p_8) = -1$, $M(p_{12}) = -1$ and $M(p_{13}) = -1$, the corresponding MFs are n_3, n_7, n_8, n_{12} and n_{13}. Lastly, $M(p_4) = 0$, $M(p_6) = 0$, $M(p_9) = 0$ and $M(p_{10}) = 0$, as the MFs n_4, n_6, n_9 and n_{10} are unknown.

Note. We introduce two types of MFs: measured and observed. The measured membership functions are either obtained experimentally or assigned by a subject expert. We denote measured MF of proposition d_i:

$$n_i = \mu_{A_i}(x_i) \quad \text{for all } i,$$

and use these MFs for the construction of fuzzy relations. We further denote observed MFs $n'_i = \mu_{A'_i}(x_i)$, if $M(p_i) = +1$ and $n'_i = \mu_{\bar{A}'_i}(x_i)$ if $M(p_i) = -1$ for places with tokens. We here briefly outline the principle of FLN construction following [21].

1. Let $d_j = x_j$ is A_j, for $j = 1$ to n be n fuzzy propositions. Let R_i be a FPR of the format: *if* d_1 *and* d_2 *and* ... *and* d_k *Then* d_{k+1} *or* d_{k+2} *or* ... *or* d_{k+n}. We represent the rule by a transition tr_i having input places $p_1, p_2, \ldots, p_k$ and output places $p_{k+1}, p_{k+2}, \ldots, p_{k+n}$, where proposition d_j is associated with place p_j, $\forall j$. We now assign a MF n_j to the place $d_j \exists j$.
2. For each pair of fuzzy propositions $d_j = x_j$ is A_j and $d'_j = x_j$ is A'_j where d_j appears in the consequent of FPR R_i and d'_j appears in the antecedent of FPR R_k, we define a place p_j as the output place of transitions tr_i and also as the input place of transition tr_k, i.e., $p_j \varepsilon \{0(\text{tr}_i) \cap I(\text{tr}_k)\}$, satisfying the FPRs R_i and R_k respectively, and associate the fuzzy proposition $d'_j = x_j$ is A'_j to place p_j.
3. Use steps 1 and 2 repeatedly so as to transform all the given rules into a FLN.

We now mention below the enabling and firing conditions of a transition.

Definition 2. A transition tr_i is **enabled** if all excluding one proposition associated with transition tr_i have tokens and the sign of the token for propositions at input (output) places of tr_i is positive (negative).

Definition 3. An enabled transition **fires** by generating MF (symbolized by token) at the place associated with the transition without any token. However, unlike traditional Petri Nets [9], the tokens are not removed from the input (output) places of the transitions after forward (backward) firing. The preservation of token at places associated with a fired transition is required for subsequent firing of other rules sharing those as input/output places. It may be added here that a transition is disabled permanently after firing once.

It is apparent from the definition of fuzzy forward chaining that an enabled transition can fire by producing a new token (and hence MF) at the desired output place. The computation of MF at the output place is performed by transposition of fuzzy propositions with tokens from the output to the input places of the selected transition after sign reversal of their tokens (representing complementation (negation) of components of the MF) and then by using traditional fuzzy reasoning. The fuzzy reasoning involved evaluates the relational matrix R from the measured MF of the propositions located at the input places after transposition and the measured MF of the proposition located at the output place. Finally, fuzzy compositional rule of inference is used to derive the MF of the proposition located at the output place from the observed MF of the propositions located at the input place after transposition of propositions.

The definition of fuzzy backward chaining, introduced earlier, reveals that an enabled transition can fire by production of a new token at the desired input place of the fired transition. However, the computation of MF at the input place of a proposition is performed by a two step procedure. In the first step, we perform transposition of the fuzzy propositions with tokens to the input place of the transition after sign reversal of the tokens (and also complementation of the components of the MFs). We also perform sign reversed transposition of the fuzzy proposition without token from the input place to the output place. In the second step, we construct a fuzzy implication relation by taking t-norm of the measured MFs of the propositions located at the input places and that of the output place, and later use the t-norm of the observed MFs of the propositions located at the input places and use compositional rule of inference to determine the MF of the proposition without token, located at the output place.

3.1. *Algorithm for* ad hoc *reasoning in a fuzzy logic network*

Input: A fuzzy logic network and fuzzy observations with known (observed) MFs for a few propositions and measured MFs for all propositions in the network

Output: Fuzzy inferences of propositions without observed MFs (or tokens)

Begin
While Tr $\neq \varnothing$ **do** // Tr = set of transitions//
Begin
For each transition $\text{tr}_i \in$ Tr, set of transitions
If all excluding one proposition associated with transition tr_i have tokens and the sign of tokens for propositions at input (output) places is positive (negative)
Then the transition is enabled.
End-For;
For each enabled transition $\text{tr}_i \in$ Tr **do**
Begin
//Generate MF for the proposition without token//

If only one output place q of tr_i does not have token **Then do**
Begin

1. **If** tr_i has more than one output places **Then** take the t-norm of the observed MFs for the propositions associated with input places and the negated MFs obtained from all excluding one proposition q located at the output places with tokens;
2. Obtain Diens–Rescher type fuzzy implication relational matrix R with antecedent MF = t-norm of the measured MFs for the propositions associated with input places and negated (measured) MFs for the propositions obtained from the output places with tokens, and consequent MF = the measured MF for the output proposition q without token, where the implication relation is (1-antecedent MF) s (consequent MF).
3. MF of the output proposition q = [t-norm obtained in step 1] $\circ$ R, where $\circ$ is the max-min composition operator;

End
Else If only one input place q' of tr_i does not have token **Then do**
Begin

4. **If** tr_i has one or more output places **Then** take the t-norm of the observed MFs for the propositions associated with input places with tokens and the negation of the measured MFs obtained for all the propositions located at the output places;
5. Obtain Diens–Rescher type fuzzy implication relational matrix R with antecedent MF = t-norm of the negated (measured) MFs for the propositions associated with the output places and propositions with known MFs obtained from the input places, and consequent MF = the negation of the measured MF for the proposition q' associated with the input place without token;
6. Negated MF of the proposition located in the input place without tokens = [t-norm obtained in step 4] $\circ$ R, where $\circ$ is the max-min composition operator;
7. Complement the obtained MF of the fuzzy proposition located at input place q' of the transition tr_i;

End-If;
Tr = Tr − {tr_i}; //Drop the transition which has fired to generate tokens at one input/output place//
End-For;
If no transition is enabled exit;
End-While;
End.

3.2. Illustrating the ad hoc reasoning algorithm

Let
d_i be a fuzzy proposition, representing x_i is A_i; each d_i is associated with a given place p_i;
$\mu_{A_i}(x_i)$ be the measured MF associated with the fuzzy proposition: x_i is A_i.
We now list the FPRs to be used in the present reasoning problem.
R_1: If d_1 and d_2 Then d_3 or d_4.
R_2: If d_4 and d_5 Then d_6 or d_7.
R_3: If d_6 and d_9 Then d_8 or d_{10}.
R_4: If d_{10} and d_{11} Then d_{12} or d_{13}.

Here, measured MFs are given at all the places p_i corresponding to the propositions $d_i \varepsilon \{d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8, d_9, d_{10}, d_{11}, d_{12}, d_{13}\}$. These MFs are required to construct the fuzzy implication relations to be used for subsequent reasoning. Moreover, the observed MFs are available for propositions with solid dots at their respective places p_1, p_2, p_5, and p_{11} and void dots at places p_3, p_7, p_8, p_{12} and p_{13}. We need to generate inferences for the propositions d_4, d_6, d_9, and d_{10} with no tokens (dots) at their corresponding places $p_j \varepsilon \{p_4, p_6, p_9, p_{10}\}$.

It is apparent from Fig. 2(a) that transitions tr_1 and tr_4 are enabled as for both the transitions all excluding only one place have tokens (representing the existence of observed MFs). So, both the transitions fire concurrently, following the *ad hoc* reasoning algorithm. Now, on firing of tr_1, MF (and hence token) is produced for proposition d_4 at place p_4 (Fig. 2(b)). Again, on firing of tr_4, the MF (and hence token) is generated for proposition d_{10} at place p_{10} (Fig. 2(b)). Next tr_2 is enabled as all excluding one input/output place of tr_2 have tokens; so it is ready for firing. Now, on firing of tr_2, token (and MF) is generated for proposition d_6 at place p_6 (Fig. 2(c)). Lastly, transition tr_3 is enabled as all excluding one input/output place of tr_3 have tokens. So, it is fired and a new token (and also MF) is generated at place p_9 for proposition d_9 (Fig. 2(d)).

The summary of the results of transition firing and MFs generated using the desired implication relations are listed sequentially in Table 1. It is apparent from the table that transitions tr_1 and tr_4 fire concurrently, following which transitions tr_2 and tr_3 fire in sequence. The notation used in the implication relations of Table 1 is apparent from its usual viewpoints.

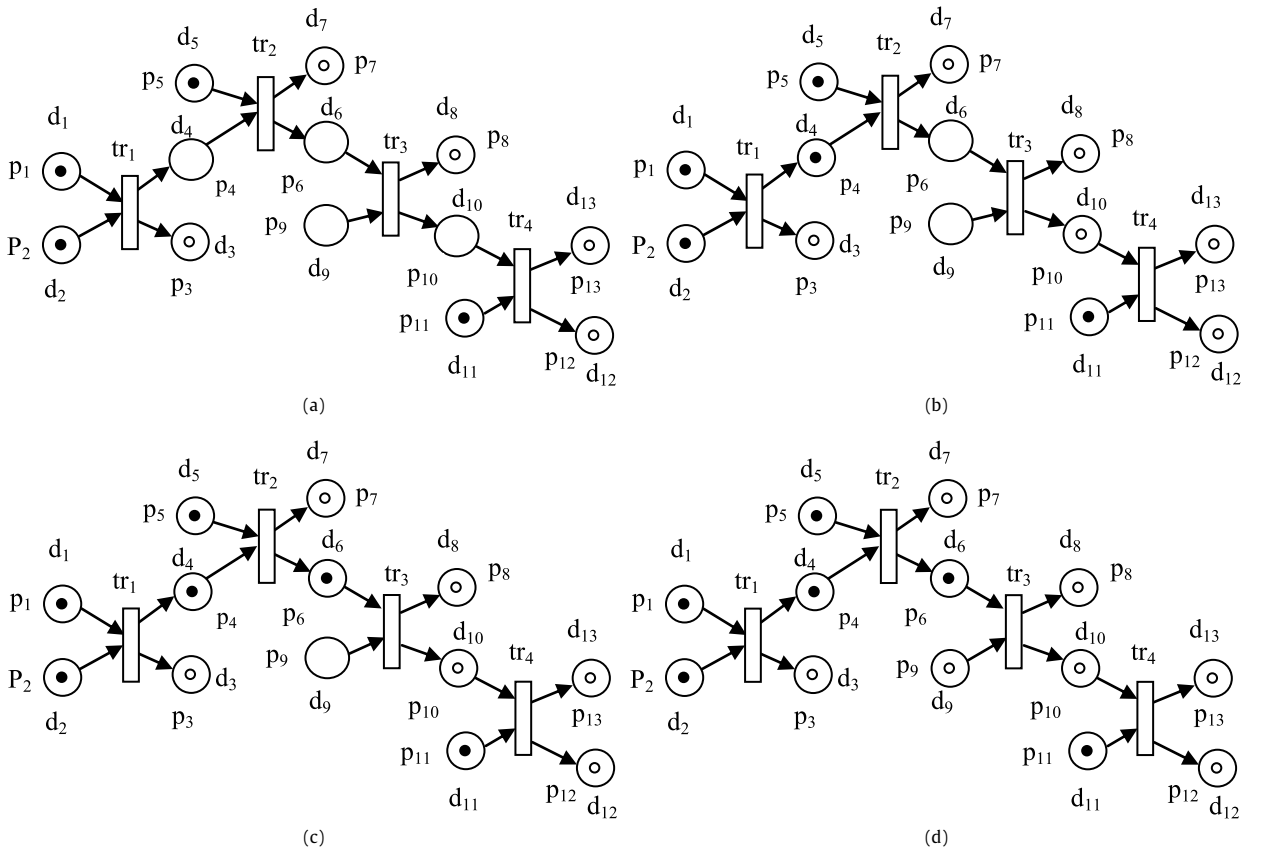


Fig. 2. (a) Initial FLN with assignment of observed MF as tokens at few places; solid dots denote that propositions are true with certain degree of membership, while void dots denote that propositions are false with certain degree of membership. Here, transitions tr_1 and tr_4 are enabled but did not fire yet. (b) After firing of tr_1 and tr_4. Here tr_2 is enabled but not fired yet. (c) After firing of tr_2. Now tr_3 is enabled but not fired. (d) After firing of tr_3, the last negative token (membership) is generated at place p_9.

Table 1
Inferences derived using selected implication relations.

Fired transitions tr_i	Implication relations constructed	Inferences derived
tr_1	$R_{d_1,d_2,\neg d_3\to d_4}$ $= (\neg(\mu_{A_1}(x_1)\,\mathrm{t}\,\mu_{A_2}(x_2)\,\mathrm{t}\,\mu_{\bar{A}_3}(x_3))\,\mathrm{s}\,(\mu_{A_4}(x_4))$	$[\mu_{A'_4}(x_4)]$ $= [(\mu_{A'_1}(x_1)\,\mathrm{t}\,\mu_{A'_2}(x_2)\,\mathrm{t}\,\mu_{\bar{A}'_3}(x_3))] \circ R_{d_1,d_2,\neg d_3\to d_4}$.
tr_4	$R_{d_{11},\neg d_{12},\neg d_{13}\to\neg d_{10}}$ $= (\neg(\mu_{A_{11}}(x_{11})\,\mathrm{t}\,\mu_{\bar{A}_{12}}(x_{12})\,\mathrm{t}\,\mu_{\bar{A}_{13}}(x_{13}))\,\mathrm{s}\,(\mu_{\bar{A}_{10}}(x_{10}))$	$[\mu_{\bar{A}'_{10}}(x_{10})]$ $= [(\mu_{A'_{11}}(x_{11})\,\mathrm{t}\,\mu_{\bar{A}'_{12}}(x_{12})\,\mathrm{t}\,\mu_{\bar{A}'_{13}}(x_{13}))] \circ R_{d_{11},\neg d_{12},\neg d_{13}\to\neg d_{10}}$; $\mu_{A'_{10}}(x_{10}) = 1 - \mu_{\bar{A}'_{10}}(x_{10})$.
tr_2	$R_{d_4,d_5,\neg d_7\to d_6}$ $= (\neg(\mu_{A_4}(x_4)\,\mathrm{t}\,\mu_{A_5}(x_5)\,\mathrm{t}\,\mu_{\bar{A}_7}(x_7))\,\mathrm{s}\,(\mu_{A_6}(x_6))$	$[\mu_{A'_6}(x_6)]$ $= [(\mu_{A'_4}(x_4)\,\mathrm{t}\,\mu_{A'_5}(x_5)\,\mathrm{t}\,\mu_{\bar{A}'_7}(x_7))] \circ R_{d_4,d_5,\neg d_7\to d_6}$.
tr_3	$R_{\neg d_8,\neg d_{10},d_6\to\neg d_9}$ $= (\neg(\mu_{\bar{A}_8}(x_8)\,\mathrm{t}\,\mu_{\bar{A}_{10}}(x_{10})\,\mathrm{t}\,\mu_{A_6}(x_6))\,\mathrm{s}\,(\mu_{\bar{A}_9}(x_9))$	$[\mu_{\bar{A}'_9}(x_9)]$ $= [(\mu_{\bar{A}'_8}(x_8)\,\mathrm{t}\,\mu_{\bar{A}'_{10}}(x_{10})\,\mathrm{t}\,\mu_{A'_6}(x_6))] \circ R_{\neg d_8,\neg d_{10},d_6\to\neg d_9}$; $\mu_{A'_9}(x_9) = 1 - \mu_{\bar{A}'_9}(x_9)$.

4. Reorganizing FLN for forward reasoning

In this section, we propose an algorithm to transform a given FLN, requiring both forward and backward firing of transitions, into an equivalent FLN, requiring forward reasoning only to determine the MF of the fuzzy propositions without tokens. The algorithm includes 6 steps. In the first step, we identify the enable transitions in the FPN. This is done by checking whether all but one place associated with the transitions have properly signed tokens. In the second step, we check whether the enabled transitions are fire-able backwards. If so, we reconfigure them into equivalent forward-firing transitions. In the third step we fire the enabled transitions in parallel and compute MFs at the output place of the transitions without token. If the resulting token appears in complemented form, re-complement it in the fourth step. This might

happen if one input place was empty in the given FLN and thus the proposition associated with the input place appears in complemented form at an output place. If no transition is enabled, we exit from the algorithm in step 5. Otherwise in step 6 we drop the fired transition.

Algorithm for forward reasoning in an equivalently transformed FLN
Begin
While Tr $\neq \varnothing$ **do** // Tr = set of transitions//
Begin
1. **For** each transition tr_i, $i = 1$ to n
   If all excluding one input/output place of the transition have properly signed tokens (i.e., negative tokens for output places and positive tokens for input places), the transition is enabled.
   End-For;
2. **For** each enabled transition tr_i,
   If the transition is fire-able backwards (i.e., only one input place does not have token and the remaining input/output places have properly signed tokens) **Then do**
   Begin
   Reconfigure the transition into its equivalent forward firing mode by the following steps;
   a) Negate the propositions located at the output places of the transition with token and connect them as input places of the transition;
   b) Negate the proposition present as an input place of the transition without token and re-define it as an output place of the transition.
   c) If $\vdash d_j$ is the resulting consequent of the fire-able transition and p_j is the output place of a second non-enabled transition then redefine p_j as the input place of the second transition with $\vdash d_j$ as the resulting proposition for place p_j. // Note: in case p_j is an input place of a second transition, then it is not enabled for firing. //
   End-If;
   End-For;
3. **For** each enabled transition tr_i //Forward pass//
   If only one output place does not have token (i.e., the transition is fire-able forward) **Then do**
   Begin
   a) **If** tr_i has more than one output places **Then** take the t-norm of the observed MFs for the propositions associated with input places and the negated MFs obtained from all excluding one proposition located at the output places with tokens;
   b) Obtain Diens–Rescher type fuzzy implication relational matrix R with antecedent MF = t-norm of the measured MFs for the propositions associated with input places and negated MFs for the propositions obtained from the output places with tokens, and consequent MF = the measured MF for the output proposition without token, where the implication relation is [(1-antecedent MF) s (consequent MF)].
   c) MF of the output proposition = [t-norm obtained in step 3a] $\circ$ R, where $\circ$ is the max-min composition operator.
   End-If;
   End-For;
4. **If** the resulting inferential propositions appear in the complemented form, re-complement it to get the final inference.
5. **If** no transition is enabled exit;
6. Tr $\leftarrow$ Tr $-$ $\{tr_i\}$;
End-while;
End.

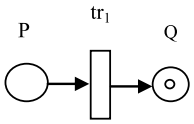
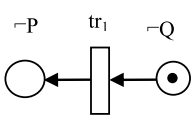
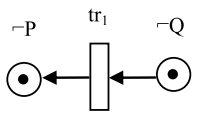
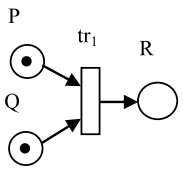
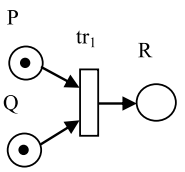
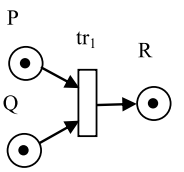
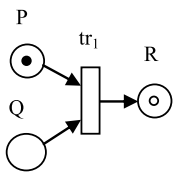
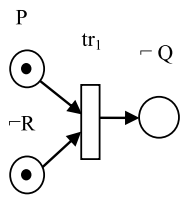
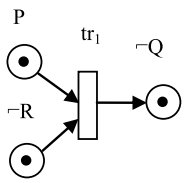
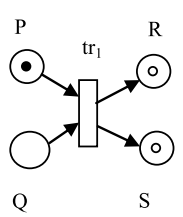
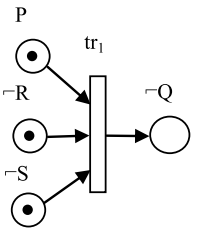
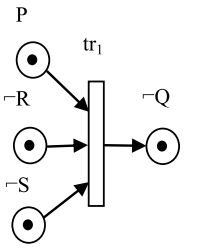
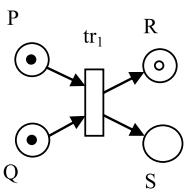
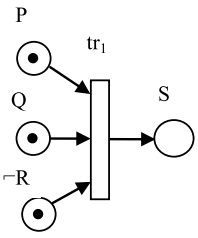
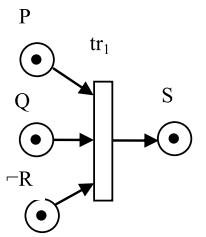
Table 2 illustrates the semantics of forward and backward firing of transitions. For backward fire-able transitions, we use steps 2(a) and 2(b) of the above algorithm. Step 2(c) is illustrated in Example 5. The table includes the representation of the given rules, token assignment following the facts, and the equivalent FLNs after using the transformation, and finally the situation after firing of the transition.

Example 5. Semantic analysis of the backward to equivalent forward transformation

This example illustrates the algorithm for forward reasoning in an equivalently transformed FLN. Fig. 3(a) indicates an FLN with tokens initialized at certain places p_1, p_2, p_7, p_8 and p_6. Let the observation at places p_1, p_2, p_7, p_8 and p_6 be x_1 is A'_1, x_2 is A'_2, y_1 is $\bar{B}'_1$, y_2 is $\bar{B}'_2$, and x_6 is A'_6. In Fig. 3(a), transitions tr_1 and tr_3 are concurrently enabled as all excluding one place associated with each of them have properly signed tokens. Since tr_1 is enabled for forward firing and only one of the output places is empty, no transformation is required for tr_1. However, for tr_3 we need to replace d_5 by an equivalent computation, shown in Fig. 4(b). In Fig. 3(b), we describe the tr_3 by its equivalence model. Now, by step 2(c) of the algorithm introduced in section 7, we transpose $d_5 = x_5$ is A_5 to the input side of the transition tr_2 after complementation of the proposition d_5.

Since tr_1 is already enabled for forward firing, we expect a positive token at place d_3. Again as tr_3 in Fig. 3(c) is enabled to produce a positive token at place p_5, tr_2 is enabled forward. Consequently, on firing of tr_2, we expect a positive token at place $\vdash d_4$. The firing order of transitions: tr_1, tr_3 first, and tr_2 later is illustrated in Fig. 5(a) and Fig. 5(b) respectively.

Table 2
Equivalent fuzzy logic networks.

Original fuzzy rule and facts	The FLN corresponding to the given rule and facts (before firing)	Equivalent FLN constructed by equivalent rule and facts (before firing of transition)	Equivalent FLN constructed by equivalent rule and facts (After firing of transition)	Equivalent rules and facts, justifying the results of firing using transformed rule
Rule : P → Q Facts: ⌐Q Inference: ⌐P				Rule : ⌐Q → ⌐P Facts : ⌐Q Inference: ⌐P (by Modus Ponens)
Rule: $P \wedge Q \rightarrow R$ Facts: P, Q Inference: R (by forward chaining)				Rule: $P \wedge Q \rightarrow R$ Facts: P, Q Inference: R (by Modus Ponens)
Rule: $P \wedge Q \rightarrow R$ Facts: P, ⌐R Inference: ⌐Q (by backward chaining)				Rule: $P \wedge \neg R \rightarrow \neg Q$ Facts: P, ⌐R Inference: ⌐Q (by Modus Ponens)
Rule: $P \wedge Q \rightarrow R \vee S$ Facts: P, ⌐R, ⌐S Inference: ⌐Q (by backward chaining)				Rule: $P \wedge \neg R \wedge \neg S \rightarrow \neg Q$ Facts: P, ⌐R, ⌐S Inference: ⌐Q (by Modus Ponens)
Rule: $P \wedge Q \rightarrow R \vee S$ Facts: P,Q,⌐R Inference: S (by forward chaining)				Rule: $P \wedge Q \wedge \neg R \rightarrow S.$ Facts: P, Q, ⌐R Inference: S (by Modus Ponens)

Theorem 4. *An ad hoc reasoning problem, capable of producing tokens at all the empty places, can equivalently be represented by a forward reasoning problem.*

Proof. An *ad hoc* reasoning problem, capable of producing tokens at the empty places, indicates that all the transitions in the network are enabled and hence fire-able at some point of time. However, a few of them may fire forward, while the rest may fire backward. Now, without changing their order of firing, we can simply represent the backward fire-able transitions into equivalent forward firing transitions. Thus all the transitions now are enabled and fire-able forward without changing

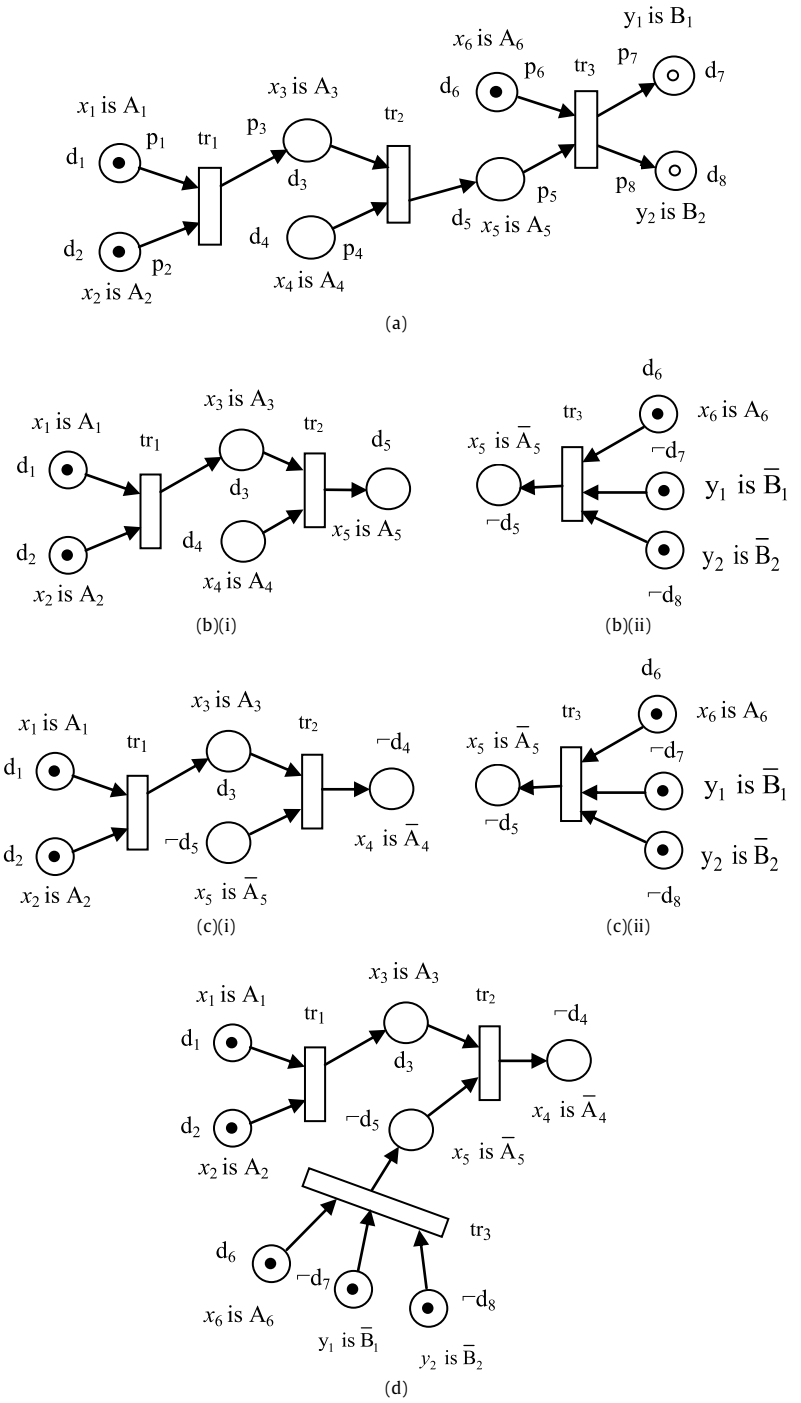


Fig. 3. (a) Original FLN with assigned tokens. (b)(i) This is part of the original Fig. 3(a). (b)(ii) This part is obtained by transformation shown Fig. 4. (c)(i) After transposition of d_4 and d_5. (c)(ii) Retained as in Fig. 3(b)(ii). (d) After combining Fig. 3c(i) and 3c(ii).

their order of firing. Consequently, the *ad hoc* reasoning problem is transformed into an equivalent forward firing model, representing a forward reasoning problem. □

5. Application in diagnosis

This section examiner the diagnosis problem of a full-wave rectifier circuit, which is often used in D.C. power supply in electronics laboratories. A schematic circuit-level diagram and meter-positioning on the circuit is shown in Fig. 6.

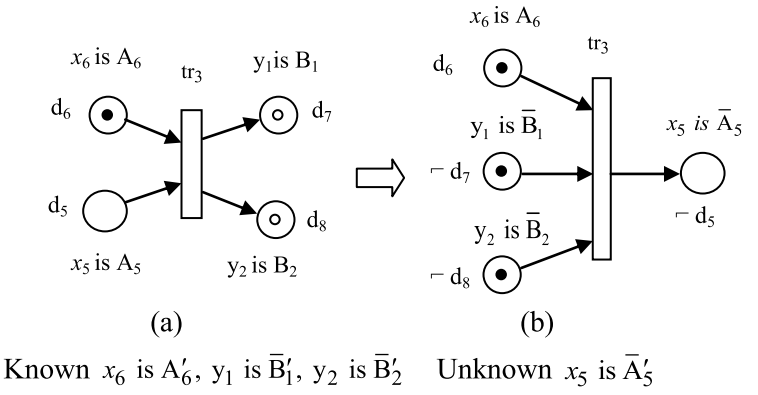


Fig. 4. Illustrating the equivalence of the FPR indicated by tr_3 in Fig. 3(a).

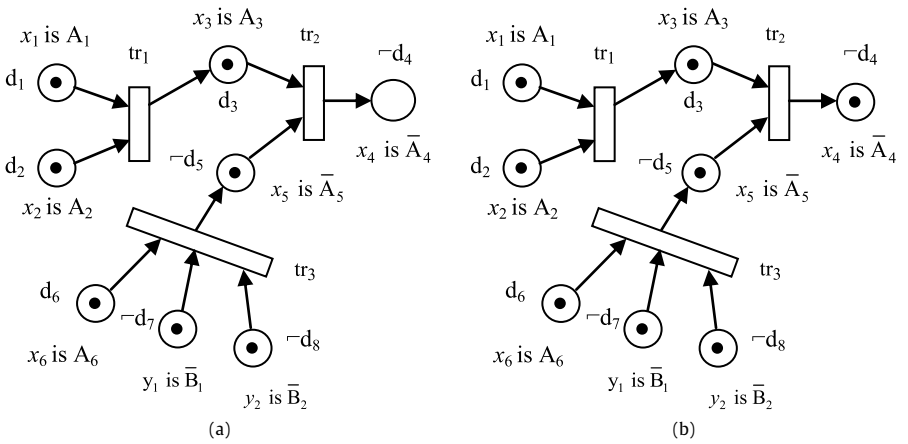


Fig. 5. (a) The FLN of Fig. 3(d) after firing of tr_1 and tr_3. (b) The FLN of Fig. 5(a) after firing of tr_2.

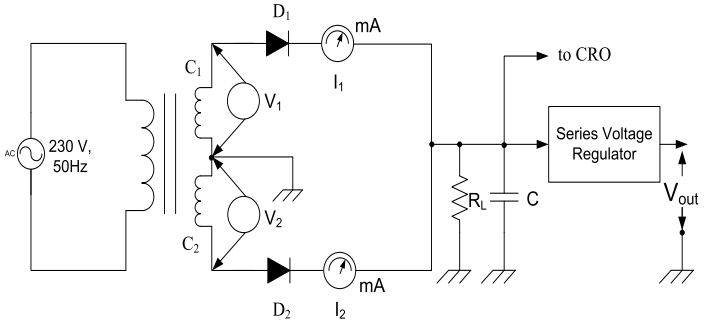


Fig. 6. Voltage and current measurement points in a full-wave rectifier circuit.

The following FPRs are used to diagnose the fault in Fig. 6.

R_1: If I_1 is NORMAL and V_{OUT} is CLOSE-TO six volts Then Coil (C_2) of the secondary transformer winding is damaged or diode D_2 is damaged.

R_2: If I_2 is NORMAL and V_{OUT} is CLOSE-TO six volts Then Coil (C_1) of the secondary transformer winding is damaged or diode D_1 is damaged.

R_3: If Coil (C_2) is damaged Then V_2 is CLOSE-TO 0 V.

R_4: If Coil (C_1) is damaged Then V_1 is CLOSE-TO 0 V.

R_5: If Diode (D_1) is damaged Then I_1 is CLOSE-TO 0 mA.

R_6: If diode (D_2) is damaged Then I_2 is CLOSE-TO 0 mA.

Fig. 7(a) indicates the mapping of the rules into FLN and initialization of tokens as observed phenomena (facts). Facts are obtained from the experimental measurements. Fig. 7(b) to (d) indicate the sequence of transition firing using the algorithm given in section 4. Table 3 provides the results of transition firing in sequence following Fig. 7(b) to (d).

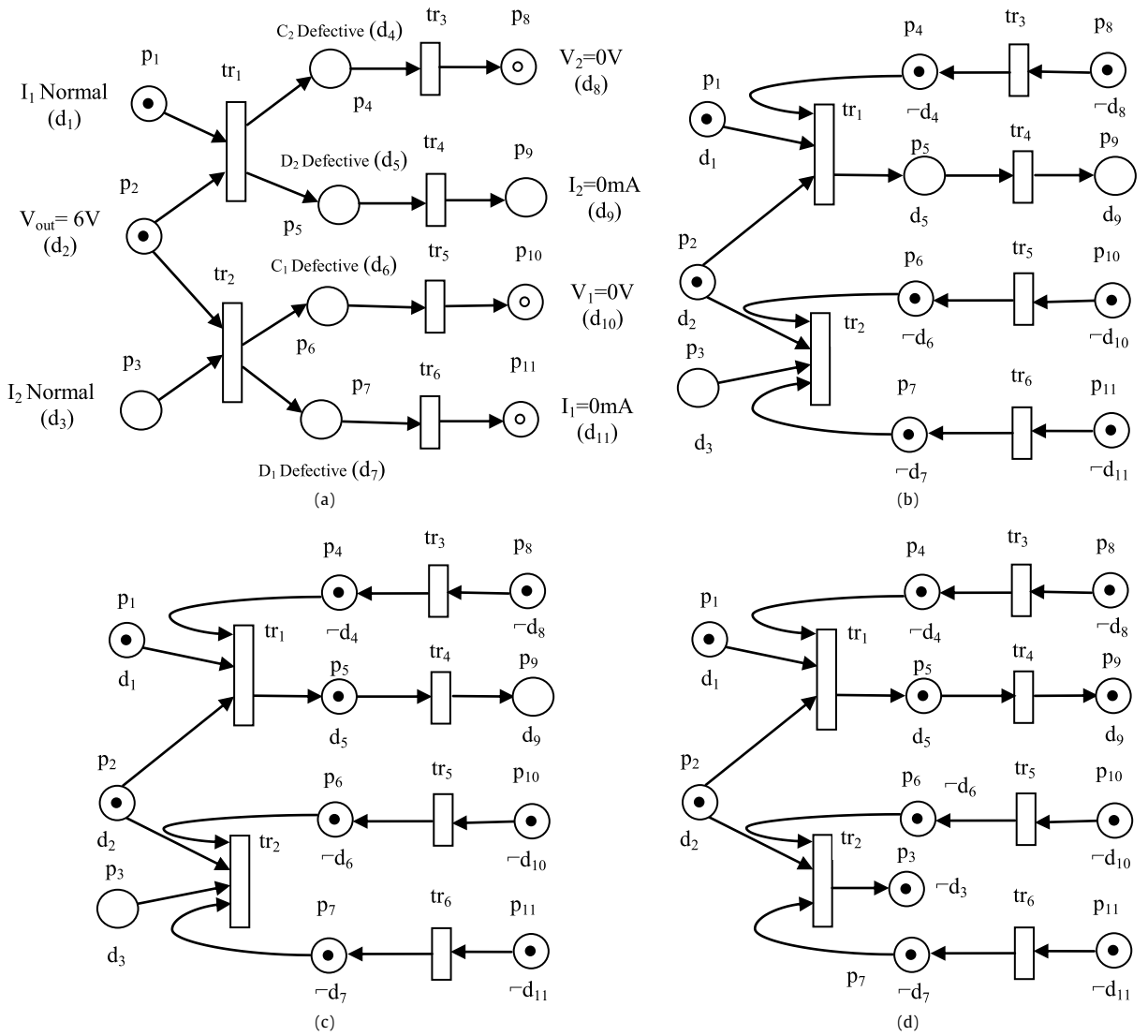


Fig. 7. (a) Initial FLN: $d_1 = I_1$ Normal, $d_2 = V_{\text{out}}$ is 6 V, $d_3 = I_2$ Normal, $d_4 = C_2$ Defective, $d_5 = D_2$ Defective, $d_6 = C_1$ Defective, $d_7 = D_1$ Defective, $d_8 = V_2$ is 0 V, $d_9 = I_2$ is 0 mA, $d_{10} = V_1$ is 0 V, $d_{11} = I_1$ is 0 mA. (b) Transitions tr_3, tr_5 and tr_6 fire with a positive token each at places p_4, p_6 and p_7. (c) Transition tr_1 fires with a positive token at place p_5. (d) Transition tr_4 fires with a positive token at place p_9 and transition tr_2 fires with a positive token at place p_3.

Table 3
Results of firing of transitions.

Fig.	Fired transitions tr_i	Inferred propositions	MF of the inferred propositions at outplace of tr_i
7(b)	tr_3	$\neg\, d_4$	$[\overline{n'_4}] = \overline{[n'_8]} \circ R(\bar{n}_8, \bar{n}_4)$;
			$[n'_4] = [1 - \overline{n'_4}]$.
	tr_5	$\neg\, d_6$	$[\overline{n'_6}] = [\overline{n'_{10}}] \circ R(\bar{n}_{10}, \bar{n}_6)$;
			$[n'_6] = [1 - \overline{n'_6}]$.
	tr_6	$\neg\, d_7$	$[\overline{n'_7}] = \overline{[n'_{11}]} \circ R(\bar{n}_{11}, \bar{n}_7)$
			$[n'_7] = [1 - \overline{n'_7}]$.
7(c)	tr_1	d_5	$[n'_5] = [(n'_1 \,\text{t}\, n'_2 \,\text{t}\, \overline{n'_4})] \circ R((n_1, n_2, \bar{n}_4), n_5)$.
7(d)	tr_2	$\neg\, d_3$	$[\overline{n'_3}] = [(n'_2 \,\text{t}\, \overline{n'_6} \,\text{t}\, \overline{n'_7})] \circ R((n_2, \bar{n}_6, \bar{n}_7), n_3)$;
			$[n'_3] = [1 - \overline{n'_3}]$.
	tr_4	d_9	$[n'_9] = [n'_5] \circ R(n_5, n_9)$.

Table 4
Measured membership functions for the propositions in Fig. 7(a).

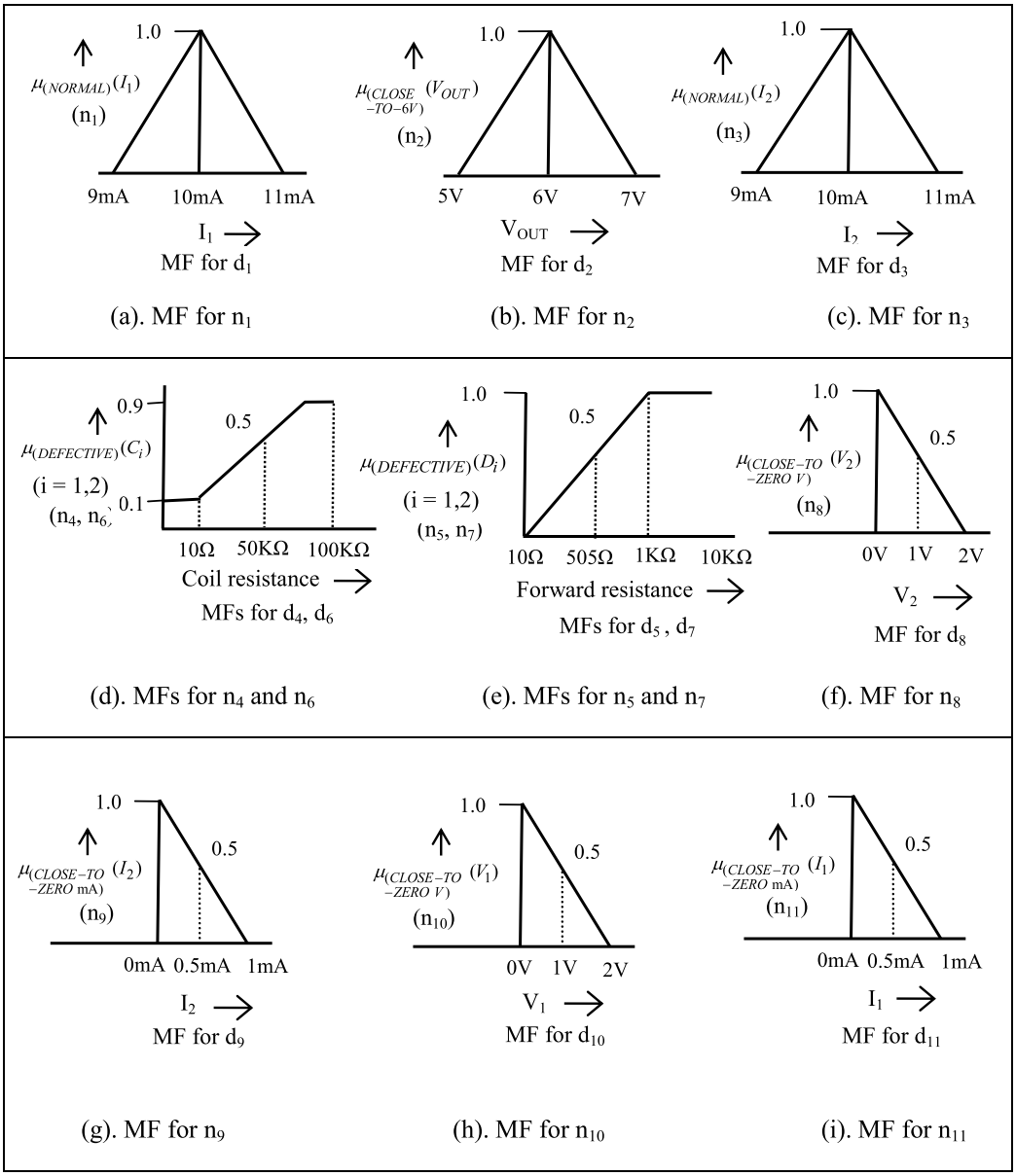


Table 4 includes the expert-defined (measured) MF for all propositions used in the FLN of Fig. 7(a). Table 5, on the other hand, provides the observed MFs for 5 propositions d_1, d_2, d_8, d_{10} and d_{12}. After reasoning is performed in Fig. 7 using the algorithm given in Section 4, we obtain the MFs for the propositions in the same figure without tokens. The resulting MFs for the propositions include d_3, d_4, d_5, d_6, d_7 and d_9.

It is apparent from Table 6 that forward resistance of diode D_2 in Table 6 being abnormally high (1K-Ohm) with a high membership of 0.9 concludes that diode D_2 is defective in the present settings.

6. Soundness and completeness issues

We begin this section with definitions of soundness and completeness [13] in the context of fuzzy reasoning and then propose two theorems to indicate that the fuzzy logic based reasoning introduced here are both sound and complete.

Table 5
Observed membership functions of propositions with tokens in Fig. 7(a).

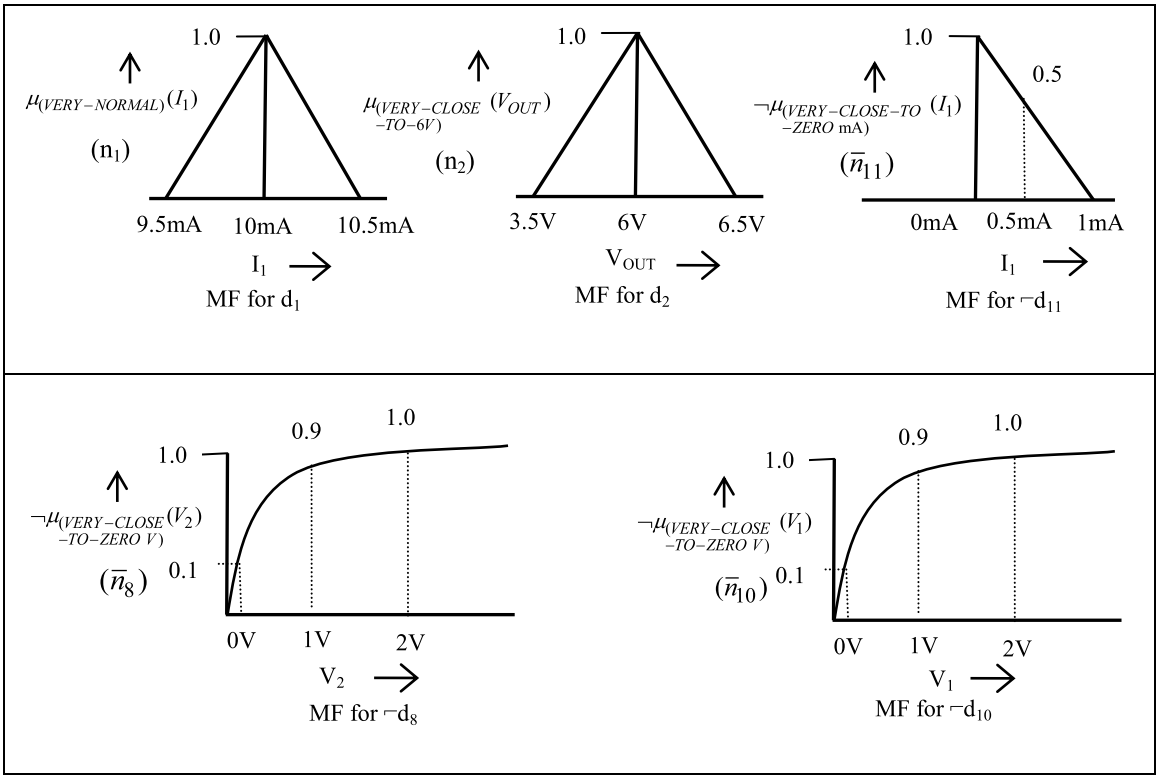


Table 6
Inferred MFs for the propositions without tokens.

Coil Resistance (C_2)	10 Ω	50 KΩ	100 KΩ
$[\overline{n'_4}] =$	[0.9	0.5	0.5]
Coil Resistance (C_1)	10 Ω	50 KΩ	KΩ
$[\overline{n'_6}] =$	[0.9	0.5	0.5]
Forward Resistance (D_1)	10 Ω	505 Ω	1 KΩ
$[\overline{n'_7}] =$	[0.5	0.5	0.5]
Forward Resistance (D_2)	10 Ω	505 Ω	1 KΩ
$[n'_5] =$	[**0.6**	**0.6**	**0.9**]
I_2 Normal	9 mA	10 mA	11 mA
$[\overline{n'_3}] =$	[0.5	0.5	0.5]
$I_2 = 0$ mA	0 mA	0.5 mA	1 mA
$[n'_9] =$	[0.9	0.6	0.6]

Definition 4. A fuzzy logic based reasoning procedure is **sound** if the set of inferences D' derived by that procedure semantically follows from the given set of FPRs R and a set of fuzzy propositions D with known MFs (or tokens in the context of FLN), i.e. $R \cup D \vdash D'$ holds.

Definition 5. A fuzzy logic based reasoning procedure is **complete** if the set of inferences D' derived by that procedure using a given set of FPRs R and a set of fuzzy propositions D with known MFs (or tokens in the context of FLN) is equal to (and not a subset of) the set of the inferences semantically derivable from R and D.

Theorem 5. *The equivalent forward firing model of an ad hoc reasoning problem is complete.*

Proof. Let $S = \{R_1, R_2, ..., R_m\}$ and $D = \{d_1, d_2, ..., d_l\}$ be the set of FPRs and fuzzy propositions with tokens, the latter being used for instantiation of the fuzzy propositions used in the rules. Let $D' = \{d_{l+1}, d_{l+2}, ..., d_n\}$ be the set of propositions with

no tokens. In other words, the corresponding places $\{p_{l+1}, p_{l+2}, ..., p_n\}$ are empty. So, we need to derive $\{d_{l+1}, d_{l+2}, ..., d_n\}$ from $S \cup U$, i.e., $S \cup U \vdash D'$, where $\vdash$ denotes provability of D' from $S \cup U$. Suppose, by reasoning on the forward firing model, we derive a set of fuzzy inferences $\{\delta\} \subset D'$. Now, if all transitions in the *ad hoc* reasoning model are enabled and forward/backward fire-able over different time slots, then by Theorem 4 they are also forward fire-able in the same order as happened to be in *ad hoc* reasoning model. Now, since a transition is fired only once and all the transitions are fire-able in the forward reasoning model, $\{\delta\} \not\subset D'$. Thus $\{\delta\} = D'$. Hence, $S \cup U \vdash D'$ holds, indicating that the equivalent forward firing model of an *ad hoc* reasoning problem is complete. □

Theorem 6. *The equivalent forward reasoning model of an ad hoc reasoning problem is sound.*

Proof. Let there be n places and m transitions. Suppose, the equivalent forward reasoning model is not sound, i.e., it produces one or more incorrect fuzzy inferences. Let the incorrect inferences be $D' = \{d_{l+1}, d_{l+2}, ..., d_n\}$, i.e., for a given set of FPRs $R = \{R_1, R_2, ..., R_m\}$ and fuzzy propositions $D_1 = \{d_1, d_2, ..., d_l\}$, with tokens, $D_1 \cup R \vdash D'$ follows. However, any element of D' is generated by an equivalent rule transformation and equivalent forward firing, which are essentially sound. So, $D' = \phi$. Hence, the equivalent forward reasoning model is sound. □

7. Discussion

The work presented in the paper is useful for its distributed approach to knowledge representation and reasoning in presence of a given knowledge-base with rules having conjunctive fuzzy propositions in the antecedent and disjunctive fuzzy propositions in the consequent, and a given set of facts with observed membership functions for a few propositions. The measured membership functions of all the fuzzy propositions used in the rules are known *a priori* from subjective experts. These measured membership functions are used later to determine the fuzzy relational matrices required for a given reasoning problem. It may be mentioned here that a given FLN with known membership functions for two different set of propositions can be considered as a two different reasoning problems as the sequence of transition firing would differ for each of the problems.

There exist a lot many works on fuzzy reasoning in a structured environment [4,5,7,8,11,14,21,26,48]. An exhaustive comparison of all these works is as good as writing a review paper. In Table 7, we compare the relative performance of our work with a few selected papers based on their relevance with the present work. This table is an extension of the comparative study undertaken by Shen [26]. It is apparent from the table that this work, to the best of our knowledge and belief, is the first of its kind that attempted to explore propositional syntax and semantics for their extensions in the context of fuzzy reasoning, and employed both for automatic reasoning in a distributed network environment. The table is self-explanatory and thus needs no further explanation. The important contributions of the paper are briefly summarized below.

1) The reasoning mechanism works in two phases. In the first phase semantic analysis is performed followed by syntactic analysis in the second phase.
2) The principles of propositional forward and backward chaining have been extended for fuzzy reasoning in the semantic phase of analysis. The merit of such extension lies in maintaining meaningful inferences, supported by propositional semantics. The inferences generated thus have a strong logical basis, even though membership functions are relaxed from strict binary values $\{0, 1\}$ to fuzzy intervals $[0, 1]$.
3) In the semantic phase of FLN analysis, backwardly enabled transitions are re-structured to keep the transitions in the transformed FLN only forwardly enabled and fire-able. The users thus find the network to be as if a forward fire-able network (requiring forward reasoning).
4) Extension of one fundamental property of Wang's algorithm (negation removal [1]) in fuzzy context, coupled with traditional fuzzy compositional rule of inference is used in the syntactic phase for fuzzy inference generation.

8. Conclusions

Traditional approaches to fuzzy reasoning employs limited semantics of propositional logic, and thus usually are not complete. In other words, by traditional GMP, GMT and GHS, we only derive a subset of the semantically plausible inferences. The proposed method of fuzzy reasoning overcomes this limitation (as evident from the proof of completeness), and the inferences derived by the proposed method are also sound (i.e., they semantically follow from the given set of rules and fuzzy propositions with observed MFs (propositions with tokens at their respective places in the FLN).

Reasoning in the fuzzy logic network here is performed in two primary phases. First, we undertake semantic reasoning using a framework similar to propositional reasoning, conditioned by the existence of signed tokens at the input and the output places of a given transition to satisfy its enabling condition. An enabled transition indicates a preparatory stage to test the feasibility of semantics of a properly instantiated rule for firing. During firing stage of a transition, we first need to transform a backward fire-able transition into its equivalent forward fire-able model, and then use forward reasoning model (GMP with multiple antecedents) to derive the fuzzy inferences. In case the transition remains in forward firing mode, we simply can fire it and derive an inference by GMP with multiple antecedents.

Table 7
Comparative study of relevant existing work with the present work.

Characteristics	Chen et al. [4,5]	Looney [8]	Scarpelli et al. [14]	Yeung et al. [48]	Pedrycz and Gomide [11]	Shen [26]	Konar et al. [7]	Chakraborty et al. [21]	Janarthanan et al.
Chaining Technique	Forward and backward separately	Forward	Forward and backward	Backward	Forward	Forward	Forward	Backward	Mixed forward and backward
Level of Reasoning	One	One	Multiple	Multiple	Multiple	One	One	One	Multiple
Structure of the reasoning network	Acyclic/Cyclic	Acyclic/Cyclic	Acyclic	Acyclic	Acyclic	Acyclic	Different approaches for Acyclic and Cyclic Reasoning	Acyclic	Acyclic
Types of Fuzzy sets used	Interval value	Point value	Point value	Point value	Interval value	Interval or point value	Point value	Interval value	Interval value
Use of Adjacent places for Reasoning	Required	Not required	Required	Required	Not required	Required	Required	Not required	Not required
Data Structure	Complex	Compact	Compact	Complex	Compact	Compact	Compact	Compact	Compact
Driven by	Given facts	Given facts	Given facts	Goal or conclusion	Given facts	Given facts	Given facts	Given facts	Given facts
Mechanism of inference generation	Consequent value = antecedent value $*$ certainty	Fuzzy matrix max-min product	Fuzzy relational matrix	Degree of subsethood	Fuzzy t- and s-norms	Fuzzy relational matrix	Fuzzy matrix max-min product	Fuzzy relational matrix	Fuzzy relational matrix
Novelty of the work	Sprouting tree for forward reasoning and AND-OR tree for backward reasoning	Fuzzy truth token adaptation by recurrent equation	knowledge based intelligent search to gather relevant information to questions and query	Multi-level weighted fuzzy reasoning	Machine learning for weight adaptation in the FLN	Selection of different aggregation operations or implication functions for different rules	Intelligent search strategies for adjacent place finding and belief revision	Extension of contraposi-tion based reasoning for abduction	Extension of propositional syntax and semantics for composite forward and backward reasoning in FLN as and when required

The justification of the soundness of the proposed forward firing model is apparent as it maintains the same firing order of transitions as happens to be in the mixed forward/backward firing model just by replacing the backward firing model by its equivalent forward model only. In addition, such transformation yields the same inference at the desired place without token as happens to be in case of backward firing of the transition. The above two issues jointly ensure the soundness of the equivalent forward firing model of the FLN.

The proposed method of automated reasoning can be used in many real-world reasoning problems including diagnosis, rule-based criminal investigation, geological/geographical explorations and many other expert systems where traditional methods fail for their semantic incompleteness. The illustrative case study on diagnosis of a rectifier circuit signifies the importance of the proposed technique in complex reasoning systems requiring both forward and backward firing of transitions in a reasoning problem, which can be taken care of by the proposed technique.

The only limitation of the proposed reasoning technique is the additional time required for online re-structuring of the FPRs, to satisfy the necessary instantiation/refutation requirement of the FPRs with the available facts, and also the time required for the construction of implication relation of the re-structured rules. However, this additional cost helps in deriving semantically meaningful inferences, which could not be obtained by traditional approaches with the same knowledge-base and facts.

Future research direction includes extending the proposed FLN model for real-time applications, such as process control and sensory-motor coordination in robotics. This can be realized by associating the notion of real time with the tokens appearing at the places and also with the transitions. The time-stamps associated with the tokens would denote the occurrence of the propositions/events and the time associated with transitions would represent their effective response delay from the onset of firing. Such extension of FLN would offer the joint benefits of uncertainty handling along with timing control in complex real-time systems.

Type-2 fuzzy extension of FLN is another important open research arena, where the reasoning needs to be undertaken using interval and/or general type-2 fuzzy sets. Such extension would offer automatic reasoning with a type-2 fuzzy inference engine, realized with FLN. Finally, most of the research on FPNs is restricted on acyclic network. In [7], Konar and Mandal proposed a recurrent (cyclic) model of FPN and used a state equation to update MFs of the propositions located at the places. The model proposed in [7] can handle singleton MFs. One attractive research problem is to develop the state equation for a recurrent FPN model with multi-point MFs. The analysis of such model would offer a generic condition of stability of the dynamical recurrent FLN model.

Acknowledgement

The authors gratefully acknowledge the funding they received from the UGC funded project UGC/UPE-II in Cognitive Science (Grant no. UPE-II/COGSC/2016).

References

[1] E.R. Dougherty, C.R. Girardina, Mathematical Methods for Artificial Intelligence and Autonomous Systems, Prentice-Hall, Englewood Cliffs, NJ, 1989.
[2] E.A. Bender, Mathematical Methods in Artificial Intelligence, IEEE Computer Society Press Silver Spring, MD, 1996.
[3] A.J. Bugarin, S. Barro, Fuzzy reasoning support by Petri nets, IEEE Trans. Fuzzy Syst. 2 (2) (May 1994) 135–150.
[4] S.M. Chen, Fuzzy backward reasoning using fuzzy Petri nets, IEEE Trans. Syst. Man Cybern., Part B, Cybern. 30 (6) (Dec. 2000) 846–855.
[5] S.M. Chen, J.S. Ke, J.F. Chang, Knowledge representation using fuzzy Petri nets, IEEE Trans. Knowl. Data Eng. 2 (3) (Sept. 1990) 311–319.
[6] M.L. Garg, S.I. Ahson, P.V. Gupta, A fuzzy Petri net for knowledge representation and reasoning, Inf. Process. Lett. 39 (Aug. 1991) 165–171.
[7] A. Konar, A.K. Mandal, Uncertainty management in expert systems using fuzzy Petri nets, IEEE Trans. Knowl. Data Eng. 8 (1) (1996).
[8] C.G. Looney, Fuzzy Petri nets for rule-based decision making, IEEE Trans. Syst. Man Cybern. 18 (1) (Jan–Feb 1988).
[9] T. Murata, Petri nets: properties, analysis and applications, Proc. IEEE 77 (4) (April 1989) 541–580.
[10] J. Pearl, Distributed revision of composite beliefs, Artif. Intell. 33 (1987) 173–215.
[11] W. Pedrycz, F. Gomide, A general fuzzy Petri net model, IEEE Trans. Fuzzy Syst. 2 (4) (Nov. 1994) 295–301.
[12] W. Pedrycz, F. Gomide, An Introduction to Fuzzy Sets: Analysis and Design, MIT Press, Cambridge, Massachusetts, 1998.
[13] S. Russel, P. Norvig, Artificial Intelligence, Prentice-Hall, Englewood-Cliffs, NJ, 1995.
[14] H. Sacrpelli, F. Gomide, R.R. Yager, A reasoning algorithm for high-level fuzzy Petri nets, IEEE Trans. Fuzzy Syst. 4 (3) (Aug. 1996).
[15] H. Wang, C. Jiang, S. Liaho, Concurrent reasoning of fuzzy logical Petri nets based on multi-task schedules, IEEE Trans. Fuzzy Syst. 9 (3) (June 2001).
[16] S.K. Yu, Knowledge representation and reasoning using fuzzy Pr/T net systems, Fuzzy Sets Syst. 75 (1995) 33–45.
[17] J.S.R. Jang, C.T. Sun, E. Mizutani, Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence, Prentice-Hall International, Inc., 1997.
[18] Li-Xin Wang, A Course in Fuzzy Systems and Control, Prentice-Hall, NJ, 1997.
[19] A. Konar, Artificial Intelligence and Soft Computing: Behavioral and Cognitive Modeling of the Human Brain, CRC Press, Boca Raton, Florida, 1999.
[20] A. Konar, Computational Intelligence: Principles, Techniques and Applications, Springer, Heidelberg, 2005.
[21] A. Chakraborty, A. Konar, N.R. Pal, Lakhmi C. Jain, Extending the contraposition property of propositional logic for fuzzy abduction, IEEE Trans. Fuzzy Syst. 21 (4) (Aug. 2013).
[22] R. Janarthanan, Aruna Chakraborty, Amit Konar, Atulya K. Nagar, Ad hoc reasoning in chained fuzzy systems realized with Diens–Rescher implication, in: Proc. of Fuzz – IEEE, 2013, pp. 1–6.
[23] H.C. Liu, L. Liu, Q.L. Lin, N. Liu, Knowledge acquisition and representation using fuzzy evidential reasoning and dynamic adaptive fuzzy Petri nets, IEEE Trans. Cybern. 43 (3) (2013) 1059–1072.
[24] A. Konar, U.K. Chakraborty, P.P. Wang, Supervised learning on a fuzzy Petri net, Inf. Sci. 172 (2005) 397–416.
[25] A. Konar, U.K. Chakraborty, Reasoning and unsupervised learning in a fuzzy cognitive map, Inf. Sci. 170 (2–4) (2005) 419–441.
[26] V.R.L. Shen, Knowledge representation using high-level fuzzy Petri nets, IEEE Trans. Syst. Man Cybern., Part A, Syst. Hum. 36 (6) (2006) 1220–1227.
[27] J. Sun, S.Y. Qin, Y.H. Song, Fault diagnosis of electric power systems based on fuzzy Petri nets, IEEE Trans. Power Syst. 19 (4) (2004) 2053–2059.

[28] C. Zhou, Z. Jiang, Fault diagnosis of TV transmitters based on fuzzy Petri nets, in: IMACS Multi Conference on Computational Engineering in Systems Applications, IEEE, Oct. 2006, pp. 2003–2009.
[29] L. Gniewek, Sequential control algorithm in the form of fuzzy interpreted Petri net, IEEE Trans. Syst. Man Cybern. Syst. 43 (2) (2013) 451–459.
[30] H. Hu, Z. Li, A. Al-Ahmari, Reversed fuzzy Petri nets and their application for fault diagnosis, Comput. Ind. Eng. 60 (4) (2011) 505–510.
[31] M. Gao, M. Zhou, X. Huang, Z. Wu, Fuzzy reasoning Petri nets, IEEE Trans. Syst. Man Cybern., Part A, Syst. Hum. 33 (3) (2003) 314–324.
[32] M. Gao, M. Zhou, Y. Tang, Intelligent decision making in disassembly process based on fuzzy reasoning Petri nets, IEEE Trans. Syst. Man Cybern., Part B, Cybern. 34 (5) (2004) 2029–2034.
[33] Y. Haiwen, Y. Haibin, L. Xingshan, Fuzzy Petri nets reasoning for application of electric control system fault diagnosis, in: 2006 IEEE Conference on Robotics, Automation and Mechatronics, June 2006, pp. 1–6.
[34] Z. Ding, J. Ma, A. Kandel, Petri net representation of switched fuzzy systems, IEEE Trans. Fuzzy Syst. 21 (1) (2013) 16–29.
[35] F. Qiao, Q. Wu, L. Li, Z. Wang, B. Shi, A fuzzy Petri net-based reasoning method for rescheduling, Trans. Inst. Meas. Control 33 (3–4) (2011) 435–455.
[36] H.S. Hu, Z.W. Li, A. Al-Ahmari, Reversed fuzzy Petri nets and their application for fault diagnosis, Comput. Ind. Eng. 60 (4) (May 2011) 505–510.
[37] M.H. Ha, Y. Li, X.F. Wang, Fuzzy knowledge representation and reasoning using a generalized fuzzy Petri net and a similarity measure, Soft Comput. 11 (4) (Feb. 2007) 323–327.
[38] S.M. Chen, Weighted fuzzy reasoning using weighted fuzzy Petri nets, IEEE Trans. Knowl. Data Eng. 14 (2) (2002) 386–397.
[39] X. Li, F. Lara-Rosano, Adaptive fuzzy Petri nets for dynamic knowledge representation and inference, Expert Syst. Appl. 19 (3) (2000) 235–241.
[40] R. Yang, W.S. Leung, P.A. Heng, K.S. Leung, Improved algorithm on rule-based reasoning systems modeled by fuzzy Petri nets, in: Fuzzy Systems, 2002. Proceedings of the IEEE International Conference on FUZZ-IEEE'02, vol. 2, 2002, pp. 1204–1209.
[41] W. Pedrycz, H. Camargo, Fuzzy timed Petri nets, Fuzzy Sets Syst. 140 (2) (2003) 301–330.
[42] Z. Ding, H. Bunke, M. Schneider, A. Kandel, Fuzzy timed Petri net definitions, properties, and applications, Math. Comput. Model. 41 (2) (2005) 345–360.
[43] Z. Ding, H. Bunke, O. Kipersztok, M. Schneider, A. Kandel, Fuzzy timed Petri nets—analysis and implementation, Math. Comput. Model. 43 (3) (2006) 385–400.
[44] T. Murata, Temporal uncertainty and fuzzy-timing high-level Petri nets, in: Application and Theory of Petri Nets 1996, Springer, Berlin, Heidelberg, 1996, pp. 11–28.
[45] J. Yuan, H. Shi, C. Liu, W. Shang, Improved basic inference models of fuzzy Petri nets, in: 7th World Congress on Intelligent Control and Automation, WCICA 2008, IEEE, June 2008, pp. 1488–1493.
[46] X. Luo, M. Kezunovic, Implementing fuzzy reasoning Petri-nets for fault section estimation, IEEE Trans. Power Deliv. 23 (2) (2008) 676–685.
[47] D.S. Yeung, E.C.C. Ysang, A multilevel weighted fuzzy reasoning algorithm for expert systems, IEEE Trans. Syst. Man Cybern., Part A, Syst. Hum. 28 (2) (1998) 149–158.
[48] J. Lee, K.F. Liu, W. Chiang, Fuzzy Petri nets for modeling rule-based reasoning, Int. J. Artif. Intell. Tools 7 (4) (1998) 463–485.
[49] H.C. Liu, J.X. You, X.Y. You, Q. Su, Linguistic reasoning Petri nets for knowledge representation and reasoning, IEEE Trans. Syst. Man Cybern. Syst. 46 (4) (2016) 499–511.
[50] H.C. Liu, J.X. You, X.Y. You, Q. Su, Fuzzy Petri nets using intuitionistic fuzzy sets and ordered weighted averaging operators, IEEE Trans. Cybern. 46 (8) (2016) 1839–1850.
[51] H.C. Liu, Q.L. Lin, M.L. Ren, Fault diagnosis and cause analysis using fuzzy evidential reasoning approach and dynamic adaptive fuzzy Petri nets, Comput. Ind. Eng. 66 (4) (2013) 899–908.
[52] S.K. Sarkar, T.G. Basavaraju, C. Puttamadappa, Ad Hoc Mobile Wireless Networks: Principles, Protocols and Applications, CRC Press, 2013.
[53] J. Pearl, Probabilistic Reasoning in Intelligent System: Networks of Plausible Inference, Morgan-Kaufmann Publisher Inc., San Francisco, CA, USA, 1988.